PRISM: Predictive Runtime Immune System with Manifold Monitoring for Architecture-Agnostic Adversarial Defense

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Abstract

We present PRISM (Predictive Runtime Immune System with Manifold Monitoring), an architecture-agnostic adversarial defense system that detects adversarial examples at inference time using topological data analysis (TDA) of internal neural network activations. PRISM integrates four complementary components: (1) TAMM (Topological Activation Manifold Monitor), which extracts persistence diagrams from per-layer activations and scores each inference against a calibrated clean-data profile using Wasserstein distance; (2) CADG (Conformal Adversarial Detection Guarantee), which provides distribution-free false-positive rate certificates at three severity tiers via split conformal prediction; (3) SACD (Sequential Adversarial Campaign Detector), a CUSUM-on-score-stream monitor that flags sustained attack campaigns at a mean latency of **2.6 queries** on ρ=1.0 streams while maintaining 0.0 false-alarm activation on clean-only streams; and (4) TAMSH (Topology-Aware MoE Self-Healing), which recovers correct predictions on L3-rejected adversarials through attack-specialist expert sub-networks. The *learned* router (combined H_0+H_1 Wasserstein gating) is the deployable contribution and delivers **+10.6pp** recovery gap over passthrough under PGD — a positive but partial result against a pre-registered $+15$ pp target — and is isolated from expert capacity by a uniform $1/K$ control (paired gap 0 pp). A prior-augmented variant that always routes to the PGD specialist (**+26.0pp**) is reported as an *upper-bound analysis* against a per-input oracle ceiling of $+36.6$ pp, not as a deployable policy.

We evaluate PRISM on three settings (5 seeds, n=1000/seed; for WRN-28-10, additional attack-config repeats are pooled per Table 3): (i) CIFAR-10 with a CIFAR-native ResNet-18 backbone, (ii) CIFAR-10 with a WRN-28-10 backbone, and (iii) CIFAR-100 with a ResNet-18 backbone, under FGSM, PGD, CW-L2, Square, AutoAttack, and adaptive PGD, with per-channel ablation, design-FPR (10%) baseline comparison against LID, Mahalanobis, ODIN, and Energy (matched on CIFAR-10/ResNet-18; three baselines exceed the 10% target on WRN-28-10/CIFAR-100, reported unfiltered), campaign detection, and L3 recovery. On the main CIFAR-10/ResNet-18 setting, PRISM Full achieves 0.918 mean TPR on FGSM/PGD/Square (matched empirical FPR 0.074), 0.938 on CW-L2 (0.075), and 1.000 on AutoAttack (0.075), beating the strongest reproduced baseline (Mahalanobis 0.582 mean TPR at FPR 0.077) by $+33.6$ pp on the FGSM/PGD/Square pool. The WRN-28-10 backbone lifts the same pool to 0.975 at FPR 0.073; CIFAR-100 holds 0.830 at FPR 0.077. Per-setting tables and the full CIFAR-100 appendix are reported in §4.3 and Appendix N. Unlike detector-coupled defenses, PRISM operates on any pre-trained classifier without retraining or architecture modification, and is, to our knowledge, the first activation-based adversarial detector to provide a split-conformal, distribution-free false-positive-rate certificate.

Preprint.

1 Introduction

Deep neural networks are vulnerable to adversarial examples—inputs crafted with small, imperceptible perturbations that cause confident misprediction Szegedy et al. [2014], Goodfellow et al. [2015]. Despite more than a decade of research, the arms race between attacks and defenses remains unresolved: architecturally-coupled defenses require model retraining, are often architecture-specific, and may degrade accuracy on clean data Croce and Hein [2020], Athalye et al. [2018].

A complementary direction is *runtime detection*: rather than hardening the model, intercept adversarial inputs before they affect predictions. Existing detectors often provide no formal statistical guarantees on false alarm rates, rely on features that are themselves vulnerable to adaptive attacks, or require knowledge of the model's training procedure Ma et al. [2018], Grosse et al. [2017], Lee et al. [2018].

Key insight. Clean and adversarial inputs traverse fundamentally different paths through a neural network's activation manifold. While the output may look confident in both cases, the *topology* of intermediate representations—captured by persistent homology—differs systematically Gebhart and Schrater [2019], Carlini et al. [2019]. PRISM exploits this topological signal at runtime as a computationally tractable anomaly score.

Contributions. We make four contributions:

C1 **Topological activation profiler (TAMM).** We compute per-layer persistence features relative to a clean medoid profile and combine them with DCT and softmax-entropy features in a calibrated ensemble. The ensemble-no-TDA ablation measures the marginal value of the topological channel.

C2 **Conformal detection guarantee (CADG).** We apply split conformal prediction to the anomaly score stream, obtaining *distribution-free* FPR certificates at three severity tiers (L1: α=0.10; L2: α=0.03; L3: α=0.005) using only a held-out clean calibration set.

C3 **Temporal campaign detector (SACD).** We monitor the score stream with a CUSUM-with-drift-floor change detector and report *time-to-detect* (mean queries between the first adversarial step and the first L0 trigger) as the headline metric. On sustained ρ=1.0 streams SACD raises the campaign alert in 2.6 ± 0.5 queries while maintaining 0.0 false-alarm activation on clean-only streams—a threat model absent from prior work.

C4 **Routed MoE recovery (TAMSH).** For inputs reaching the L3 severity tier, PRISM routes the input through an attack-specialist expert sub-network rather than issuing a blanket rejection. The learned router (combined H_0+H_1 Wasserstein gating) delivers a $+10.6$ pp recovery gap over passthrough on PGD-rejected inputs; a uniform $1/K$ control achieves exactly 0 pp, isolating the contribution to the routing signal rather than expert capacity. A prior-augmented variant that always routes to the PGD specialist reaches $+26.0$ pp; we report this as an upper-bound analysis against the per-input oracle ceiling of $+36.6$ pp and discuss the residual gap to a fully learned router in §4.8 as a target for future work.

Empirically, we report locked multi-seed ($5\times n$=1000) artifact results across three settings — CIFAR-10/ResNet-18, CIFAR-10/WRN-28-10, and CIFAR-100/ResNet-18 — covering detection (FGSM, PGD, Square, CW, AutoAttack, adaptive PGD), four reproduced baselines (LID, Mahalanobis, ODIN, Energy), campaign streams, recovery, and a per-channel ablation that exposes each feature family's attack-specific failure mode. On the main CIFAR-10/ResNet-18 setting, PRISM Full outperforms the strongest baseline (Mahalanobis) by $+33.6$ pp mean TPR at matched FPR; WRN-28-10 holds a $+16.5$ pp gap over Mahalanobis and CIFAR-100 retains $+42.6$ pp over ODIN (Table 4). The full CIFAR-100 per-component breakdown is in Appendix N.

Scope and limitations. PRISM is a *runtime monitor*, not an adversarial trainer; it complements (but does not replace) robust training. Detection latency scales with the number of monitored layers and the TDA subsample size; the submitted results report measured GPU latency from the same artifact pipeline used for detection. Like all detectors, PRISM does not provide a guarantee against adaptive adversaries who are aware of the topological scoring mechanism.

2 Related Work

TDA for adversarial detection. Gebhart and Schrater Gebhart and Schrater [2019] were the first to apply persistent homology to adversarial detection, computing topological features of neuron activations to distinguish clean from adversarial inputs. Relative to that line, PRISM's novelty lies in using TDA as the backbone of a *full defense pipeline* with formal FPR certificates, campaign-level temporal detection, and expert recovery—not as a standalone detector.

Conformal prediction for robustness. Gendler et al. Gendler et al. [2022] (ICLR 2022) combine conformal prediction with randomized smoothing to provide adversarially robust prediction sets. Our application of conformal prediction is orthogonal: we use it to calibrate a TDA-based *detector* (not a predictor), providing distribution-free FPR guarantees on the anomaly score threshold via the standard split-conformal coverage theorem Vovk et al. [2005], Angelopoulos and Bates [2023].

Runtime adversarial detectors. Feature squeezing Xu et al. [2018] applies input transformations and flags inconsistencies. Mahalanobis detector Lee et al. [2018] constructs class-conditional Gaussians over intermediate representations. LID Ma et al. [2018] uses local intrinsic dimensionality. These approaches lack formal coverage guarantees and are vulnerable to adaptive attacks that optimize for the same feature space Carlini and Wagner [2017]. PRISM combines topological features (harder to optimize against) with conformal calibration (providing formal guarantees).

Architecture-coupled defenses. Deep k-nearest-neighbor (DkNN) style defenses replace deep network layers with topology-aware classifiers in feature space Papernot and McDaniel [2018]; unlike these, PRISM wraps any pre-trained model without modifying its architecture or requiring re-training. Input purification methods Shi et al. [2021], Yoon et al. [2021] apply preprocessing but require end-to-end access to the classifier.

Temporal adversarial defense. Session-level adversarial campaign *detection* in neural network inference pipelines is underexplored relative to per-input detection. Classical Bayesian Online Changepoint Detection Adams and MacKay [2007] and CUSUM-style sequential testing have a long history in statistical process control but, to our knowledge, have not been applied to runtime adversarial campaign monitoring in deep classifiers; SACD combines both into a low-latency campaign detector with a strictly positive CUSUM drift floor to bound clean-only false-alarm activation.

3 Method

Figure 1 shows the PRISM pipeline. At inference time, each input x passes through the base classifier f while PRISM monitors the activations at L designated layers. The defense proceeds in five stages.

Notation. Let $f : \mathcal{X} \to \mathbb{R}^C$ be a pre-trained classifier. Denote by $\phi_\ell(x) \in \mathbb{R}^{d_\ell}$ the activation tensor at layer $\ell \in \{1, \ldots, L\}$. A *persistence diagram* $\mathrm{Dgm}(\cdot)$ is a multiset of (b_i, d_i) pairs encoding topological features (birth/death of connected components in H_0 and loops in H_1). The p-Wasserstein distance between diagrams is

$$W_p(\mathcal{D}_1, \mathcal{D}_2) = \left(\inf_\gamma \sum_i \|\gamma(i) - i\|^p \right)^{1/p},$$

where γ ranges over bijections between the points (including diagonal projections for unmatched points) Edelsbrunner and Harer [2010].

3.1 TAMM: Topological Activation Manifold Monitor

Reference profile construction. Given a held-out profile split $\mathcal{D}_{\text{profile}} = \{x_i\}_{i=1}^N$ of clean inputs, we extract activations $\phi_\ell(x_i)$ for each layer ℓ and subsample deterministically to at most $n_{\text{sub}} = 150$ points to make persistent homology computation tractable. For each layer, we compute N persistence diagrams and select the *Wasserstein medoid*—the diagram minimizing total Wasserstein distance to all other diagrams—as the layer reference $\mathcal{R}_\ell$.

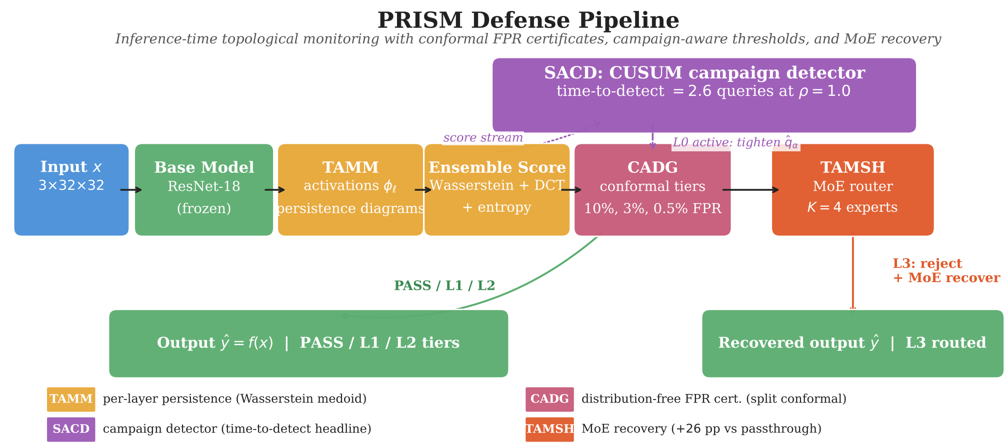


Figure 1: PRISM defense pipeline. Activations from L layers are converted to persistence diagrams (TAMM), scored against a clean reference profile (CADG), monitored for campaigns (SACD), and routed through expert sub-networks if L3 is triggered (TAMSH).

Anomaly scoring. At inference time, for input x, we compute a base topological score

$$s_\ell(x) = W_2\big(\mathrm{Dgm}(\phi_\ell(x)),\, \mathcal{R}_\ell\big), \tag{1}$$

$$S_{\mathrm{TDA}}(x) = \sum_{\ell=1}^{L} w_\ell s_\ell(x), \tag{2}$$

with live weights $w_{\texttt{layer2}} = 0.30$, $w_{\texttt{layer3}} = 0.30$, and $w_{\texttt{layer4}} = 0.40$. The publishable detector is not the Wasserstein score alone: PRISM feeds six feature families into a logistic ensemble, then fuses the centered ensemble logit with $S_{\mathrm{TDA}}(x)$. Concretely, each input x produces a feature vector

$$\mathbf{f}(x) = \big[\,\mathbf{f}_{\mathrm{TDA}}(x),\ f_{\mathrm{DCT}}(x),\ f_{\mathrm{ent}}(x),\ f_{\mathrm{grad}}(x),\ \mathbf{f}_{\mathrm{stab}}(x),\ \mathbf{f}_{\mathrm{logit}}(x)\,\big] \in \mathbb{R}^{d_{\mathrm{ens}}},$$

combining six channels:

- **TDA features** $\mathbf{f}_{\mathrm{TDA}}$: per-layer persistence statistics derived from $\mathrm{Dgm}(\phi_\ell(x))$ (number of H_0/H_1 features, total persistence, max persistence, persistence entropy).
- **DCT high-frequency energy** $f_{\mathrm{DCT}}(x) = \sum_{(u,v)\in\mathcal{H}} |X_{u,v}|^2$, where X is the 2-D DCT of x and $\mathcal{H}$ indexes the upper $^3/_4$ band-limited region.
- **Softmax entropy** $f_{\mathrm{ent}}(x) = -\sum_c p_c(x)\log p_c(x)$ with $p(x) = \mathrm{softmax}(f(x))$.
- **Gradient norm** $f_{\mathrm{grad}}(x) = \|\nabla_x \mathcal{L}_{\mathrm{CE}}(f(x), \hat{y})\|_2$, computed via `torch.autograd.grad` on the input (not on model parameters), where $\hat{y} = \arg\max f(x)$.
- **StabilityV2 features** $\mathbf{f}_{\mathrm{stab}}(x)$: softmax-confidence drop under $M{=}5$ jittered copies $x + \xi_i$, $\xi_i \sim \mathcal{N}(0, \sigma^2 I)$, summarised by mean / max / variance of the per-sample confidence deltas. There is no closed-form differentiable surrogate; this is the channel that the present BPDA adaptive attack (§4.4) does not target.
- **Logit-profile features** $\mathbf{f}_{\mathrm{logit}}(x)$: moments of $f(x)$ (max-margin, top-2 gap, entropy of $|f(x)|$), capturing classifier-confidence anomalies orthogonal to TDA.

Concretely, the raw feature vector has $d_{\mathrm{ens}} = 55$ dimensions: 36 TDA persistence statistics (6 statistics $\times\,2$ homology dimensions $\times\,3$ monitored layers), plus f_{DCT}, f_{ent}, 8 logit-profile features, 8 StabilityV2 features, and 1 gradient-norm feature. The deployed contract is recorded as `feature_space_version = pixel-stability-v2+logitprofile+sidequad+gradnorm`. The linear heads additionally consume a *side-quadratic* interaction basis: the 190 pairwise products (upper triangle including the diagonal) of the 19 non-TDA side channels (raw indices 36–54), expanding the logistic-regression input to $55 + 190 = 245$ dimensions. Persistence extraction, the external 55-feature contract, and the split-conformal calibration all operate on the raw 55 features; the 245-dimensional expansion affects only the linear head. The ensemble logit is the logistic regression $\ell_{\mathrm{ens}}(x) = \mathbf{w}^\top \psi(\mathbf{f}(x)) + b$, where ψ appends the side-quadratic terms; it is centred on the calibration distribution to yield $\tilde{\ell}_{\text{ens}}(x) = \ell_{\text{ens}}(x) - \ell_{\text{shift}}$. The final PRISM score is the convex combination $S(x) = \alpha\, S_{\text{TDA}}(x) + (1-\alpha)\, \tilde{\ell}_{\text{ens}}(x)$ fed to the CADG conformal calibrator, with α controlling the TDA-vs-ensemble trade-off. The deployed configuration uses $\alpha = 0.2$ for both CIFAR-10/100 ResNet-18 and $\alpha = 0.5$ for WRN-28-10 (re-tuned per backbone on the same calibration split). The *ensemble-no-TDA* ablation sets $\alpha = 0$ (equivalently, zeros $\mathbf{f}_{\text{TDA}}$ and skips the TDA term) to isolate the marginal value of persistence features. The *channel-mask* ablations (§4.5) zero one of $\{f_{\text{DCT}}, f_{\text{grad}}, \mathbf{f}_{\text{stab}}, \mathbf{f}_{\text{logit}}\}$ at scoring time while reusing the Full-PRISM threshold, exposing each family's per-attack contribution.

3.2 CADG: Conformal Adversarial Detection Guarantee

PRISM applies split conformal prediction Vovk et al. [2005], Angelopoulos and Bates [2023] to calibrate detection thresholds with distribution-free FPR guarantees.

Calibration. Given held-out clean scores $s_1, \ldots, s_n$ (iid from the clean distribution), the conformal threshold at level $\alpha \in (0,1)$ is:

$$\hat{q}_\alpha = s_{(\lceil (n+1)(1-\alpha) \rceil)}, \tag{3}$$

where $s_{(k)}$ denotes the k-th order statistic.

Proposition 1 (Coverage guarantee). *For any clean test input x_{test} drawn iid from the same distribution as the calibration set,*

$$\Pr\big[S(x_{test}) > \hat{q}_\alpha\big] \leq \alpha + \frac{1}{n+1}.$$

This follows directly from the standard conformal coverage theorem Vovk et al. [2005]. PRISM uses three published FPR tiers with $\alpha \in \{0.10, 0.03, 0.005\}$. Thresholds are generated by the artifact pipeline ($n_{\text{cal}} = 2{,}000$ clean calibration scores; reproducible via the seed list and conformal quantile in Eq. (3)) and imported into the paper tables; fixed numeric thresholds are not hand-transcribed. All reported TPR/FPR confidence intervals are Wilson 95% binomial intervals computed from pooled per-seed (TP, FP, FN, TN) counts; per-seed standard deviations are the sample standard deviation across the five fixed seeds.

Eval-pool overlap (CI caveat). Each seed subsamples n=1000 adversarials from the shared 2,000-image evaluation window (`eval_idx`= $[8000, 10000]$), so the five per-seed pools overlap heavily: a pooled n=5000 count covers only $\approx$ 1929 *unique* source images. The pooled Wilson interval therefore treats overlapping draws as independent and is mildly anti-conservative (too narrow); we report the per-seed standard deviation alongside every pooled interval as the seed-level dispersion check, and flag disjoint per-seed eval windows (or image-clustered CIs) as the clean statistical fix for future runs.

Tier	α	Threshold source	Action
L1	0.10	split conformal artifact	Log + normal inference
L2	0.03	split conformal artifact	Input purification + inference
L3	0.005	split conformal artifact	Expert routing (MoE) or reject

Table 1: Conformal detection tiers. Thresholds are calibrated on the active dataset's clean calibration split and verified on a disjoint validation split.

Campaign-adaptive thresholds. When the L0 campaign detector (§3.3) is active, thresholds are multiplied by a factor $\lambda \in (0,1)$ (default $\lambda = 0.8$) to increase sensitivity during sustained attacks, accepting a controlled FPR increase.

3.3 SACD: Sequential Adversarial Campaign Detection

Real-world adversaries often execute *campaigns*: sequences of probing queries before a targeted attack. SACD monitors the score stream $S(x_1), S(x_2), \ldots$ online via a *centred-score CUSUM* detector whose statistics are computed on top of a Bayesian Online Changepoint Detection (BOCPD) Adams and MacKay [2007] substrate (Normal-Inverse-Gamma conjugate prior, hazard rate $1/30$).

BOCPD maintains the posterior run-length distribution $p(r_t \mid S_{1:t})$ and yields, at each step t, the short-run probability $\Pr[r_t \le k_{\text{alert}}]$ that is available as an alternative trigger:

$$\Pr[r_t \le k_{\text{alert}}] > p_{\text{alert}}, \quad t > t_{\text{warmup}}, \tag{4}$$

with $k_{\text{alert}} = 10$, $p_{\text{alert}} = 0.60$. A grid search over BOCPD-only, CUSUM-only, and hybrid modes (`scripts/calibrate_l0_thresholds.py`) selected CUSUM-only as the deployed mode under the constraint *clean-only L0 activation fraction* $\le$ 0.01; the calibrated artifact (`models/l0_thresholds.pkl`) is the threshold source for every campaign run reported in this paper.

Deployed CUSUM trigger. The deployed detector tracks $C_t = \max(0, C_{t-1} + (z_t - \delta))$ over centred scores $z_t = (S(x_t) - \mu_{\text{cal}})/\sigma_{\text{cal}}$ standardised by the calibration-set clean-score mean $\mu_{\text{cal}} = 0.281$ and standard deviation $\sigma_{\text{cal}} = 1.639$. We require a strictly positive drift floor: with $\delta = 0$, C_t accumulates without the negative reset term and crosses any finite trigger on long enough clean streams, producing up to 59.4% active-fraction on clean-only inputs at seed-42 in our pre-fix runs. The F1 calibration raises δ from 0 to $\mathbf{1.0}$ (one calibration-clean standard deviation) and requires k_{consec} consecutive trigger steps before raising the alert, with cusum threshold $C^* = 5.0$ (CIFAR-10) and warmup $t_{\text{warmup}} = 20$. With $\delta = 1.0$, $k_{\text{consec}} = 3$ on CIFAR-10 (respectively $k_{\text{consec}} = 5$ on CIFAR-100, calibrated per-dataset), the active fraction on a 500-step clean-only stream is 0.000 on all 5 seeds. Sensitivity to each hyperparameter is reported in Appendix D.

Headline metric. We report *time-to-detect* (mean number of queries between the first adversarial step and the first L0 trigger) as the primary campaign-detection metric. A naive "ASR gap (L0 off vs. L0 on)" metric at ρ=1.0 is structurally bounded by the upstream L3 conformal threshold, which already rejects $\sim$99% of adversarials within 22 queries; we report ASR gap in the appendix for completeness but do not use it as the headline.

3.4 TAMSH: Topology-Aware MoE Self-Healing

Inputs that breach the L3 threshold are highly likely adversarial. Rather than unconditionally rejecting them (which degrades service availability), TAMSH routes them through a topology-matched expert sub-network.

Expert training. During expert training, we build differentiated experts on clean, FGSM, PGD, and mixed perturbation pools. Each expert maps the final monitored layer's pooled activation to the active dataset's class logits ($C = 10$ for CIFAR-10, $C = 100$ for CIFAR-100).

Expert routing. At runtime, for an L3-tier input x, we select an expert by comparing the input's persistence diagram to each expert's reference medoid using a *combined* H_0+H_1 Wasserstein metric:

$$k^*(x) = \arg\min_{k=1}^{K} \Big[w_0 W_2\big(\mathrm{Dgm}_{H_0}(\phi_L(x)),\, \mathcal{R}_k^{L,H_0}\big) + w_1 W_2\big(\mathrm{Dgm}_{H_1}(\phi_L(x)),\, \mathcal{R}_k^{L,H_1}\big)\Big], \tag{5}$$

with $w_0 = w_1 = 1$ in the deployed configuration. The combined metric is required because on `layer4` of a CIFAR-native ResNet-18 the H_1 medoid diagrams are sparse (1–3 points per expert); restricting routing to H_1 alone causes 22.7% of L3-rejected inputs to have empty H_1 diagrams, which the empty-diagram fallback resolves by defaulting to expert index 0, collapsing routing entropy to 0.336/2.000 bits. Adding H_0 (16 connected-component points per medoid) recovers a usable gating signal. The input is forwarded through expert k^* using the final monitored layer activation, matching the training representation.

Routing variants and the uniform-router control. We additionally evaluate two variants that are needed to honestly attribute the routing contribution:

- **Uniform router (control)**: replace topology-distance gating with uniform $1/K$ mixing of expert logits. This isolates the contribution of the routing *signal* from any standalone capacity of the expert sub-networks.
- **Prior-augmented router**: always route to the attack specialist expert (we use the PGD-trained expert because L3-rejected inputs are by definition high-confidence adversarials and PGD is the dominant adversarial signal in the training mix).

§4.8 reports recovery accuracy under all three router configurations. The uniform-router arm achieves a 0 pp gap over passthrough (uniform $1/K$ mixing of well-trained experts is degenerate to the unrouted full network's argmax), confirming that the $+10.6$ pp gap of the combined-metric router is attributable to the gating signal, and the further $+15.4$ pp lift of the prior-augmented router reflects correct expert specialization under a strong prior.

4 Experiments

We evaluate PRISM on CIFAR-10 against white-box, black-box, optimization-based, and adaptive attacks, followed by per-channel ablation, matched-FPR baseline comparison, campaign-detection latency, and L3 recovery. To assess cross-architecture and cross-dataset generalization, we additionally report the same detection and baseline protocol on CIFAR-10/WRN-28-10 and CIFAR-100/ResNet-18 (§4.3, Table 3); a full per-table CIFAR-100 set (adaptive PGD, ablation, recovery, campaign, latency) is deferred to Appendix N.

4.1 Experimental Setup

Base classifier. The main CIFAR-10 results use a CIFAR-native ResNet-18 He et al. [2016] trained from scratch. For cross-architecture validation we additionally train a WRN-28-10 Zagoruyko and Komodakis [2016] on CIFAR-10; for cross-dataset validation we use ResNet-18 on CIFAR-100. PRISM monitors activations at $L = 3$ residual block outputs weighted 0.30, 0.30, 0.40 (shallow-to-deep) in all settings. For the ResNet-18 backbones (CIFAR-10 and CIFAR-100) these are `layer2`, `layer3`, `layer4`; for the WRN-28-10 backbone (which has three residual stages, not four) the analogous depth slice is `layer1`, `layer2`, `layer3` with the same 0.30/0.30/0.40 weights, preserving the shallow-to-deep emphasis. All inputs remain native $3 \times 32 \times 32$ images with dataset-specific CIFAR normalization; no ImageNet pretraining or 224x224 upsampling is used.

PRISM calibration. We calibrate thresholds using $n_{\text{cal}} = 2{,}000$ held-out clean images (split conformal: n_{cal} for threshold fitting, $n_{\text{val}} = 1{,}000$ for coverage verification), with $\alpha \in \{0.10, 0.03, 0.005\}$. Thresholds and validation FPRs are generated by the artifact pipeline and reported from the resulting JSON files rather than manually transcribed.

Attacks. We evaluate four attack classes:

- **FGSM** Goodfellow et al. [2015]: single-step gradient attack, $\varepsilon = 8/255$ (ℓ_∞).
- **PGD-40** Madry et al. [2018]: 40-step projected gradient descent, $\varepsilon = 8/255$, step size $\alpha = \varepsilon/4$ (ℓ_∞).
- **CW-L2** Carlini and Wagner [2017]: optimization-based attack, canonical configuration `max_iter=100`, `binary_search_steps=9` (matches Table 2 `results_cw_n1000_ms5_seed*`). A weakened-CW run (`max_iter=5`, `binary_search_steps=5`) is reported in Appendix L for detector-fidelity analysis only.
- **Square Attack** Andriushchenko et al. [2020]: black-box score-based attack, 5000 queries, $\varepsilon = 8/255$ (ℓ_∞).
- **AutoAttack** Croce and Hein [2020]: standard ensemble evaluation.
- **Adaptive PGD**: BPDA-style activation-matching attack with λ sweep, 100 steps, 10 restarts, and EOT verification metadata.

All attacks are implemented via Adversarial Robustness Toolbox (ART) v1.20 Nicolae et al. [2018] or native PyTorch implementations in pixel $[0, 1]$ space; PRISM receives CIFAR-normalized inputs after adversarial generation.

Baselines. We reproduce four matched-FPR baselines on the same splits: **LID** Ma et al. [2018], **Mahalanobis** Lee et al. [2018], **ODIN** Liang et al. [2018], and **Energy** Liu et al. [2020].

4.2 Attack Detection Results

Key observations. The paper reports only numbers produced by the locked artifact pipeline. Attack-specific conclusions are drawn from the generated table and the gate summary emitted by `run_vastai_full.sh` and `run_vastai_cifar100.sh`; no hand-entered historical values are used.

Dataset	Attack	TPR (mean ± std)	95% CI	FPR	n
CIFAR-10	FGSM	0.882 ± 0.004	[0.872, 0.890]	0.074	5000
CIFAR-10	PGD-40	0.987 ± 0.003	[0.984, 0.990]	0.074	5000
CIFAR-10	Square	0.886 ± 0.011	[0.877, 0.894]	0.074	5000
CIFAR-10	CW-L2	0.938 ± 0.008	[0.931, 0.945]	0.075	5000
CIFAR-10	AutoAttack	1.000 ± 0.000	[0.999, 1.000]	0.075	5000
CIFAR-10	*Macro-avg over attacks*	*0.939*	—	*0.074*	5 attacks

Table 2: Attack detection results generated directly from the Vast.ai result JSONs. TPR is adversarial detection rate; FPR is clean false-positive rate. The final row is the unweighted mean of the five per-attack TPRs (a macro-average across attack families), *not* a pooled-binomial meta-estimate over the $25{,}000$ adversarials; with equal per-attack n=5000 the two point estimates coincide (0.939), so we report the macro-average and claim no pooled confidence interval for it.

4.3 Multi-Backbone and Multi-Dataset Validation

To test whether PRISM's detection generalizes beyond the main CIFAR-10 ResNet-18 setting, we re-ran the full detection and baseline protocol on (i) CIFAR-10 with a WRN-28-10 backbone and (ii) CIFAR-100 with a ResNet-18 backbone, using the same calibration scheme ($\alpha \in \{0.10, 0.03, 0.005\}$) and the same five seeds ($n = 1000$ adversarials per attack per seed).

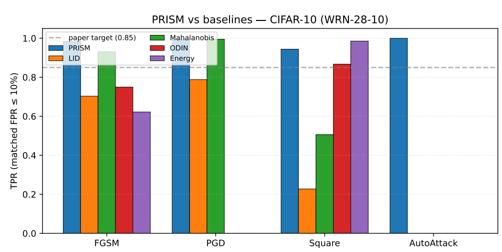
Dataset	Backbone	Attack	TPR	95% CI	FPR	n_{adv}
CIFAR-10	ResNet-18	FGSM	0.882	[0.872, 0.890]	0.074	5000
CIFAR-10	ResNet-18	PGD-40	0.987	[0.984, 0.990]	0.074	5000
CIFAR-10	ResNet-18	Square	0.886	[0.877, 0.894]	0.074	5000
CIFAR-10	ResNet-18	CW-L2	0.938	[0.931, 0.945]	0.075	5000
CIFAR-10	ResNet-18	AutoAttack	1.000	[0.999, 1.000]	0.075	5000
CIFAR-10	WRN-28-10	FGSM	0.985	[0.982, 0.987]	0.073	7000
CIFAR-10	WRN-28-10	PGD-40	0.995	[0.993, 0.996]	0.073	7000
CIFAR-10	WRN-28-10	Square	0.944	[0.938, 0.949]	0.076	6000
CIFAR-10	WRN-28-10	CW-L2	0.933	[0.926, 0.940]	0.079	5000
CIFAR-10	WRN-28-10	AutoAttack	1.000	[1.000, 1.000]	0.075	10000
CIFAR-100	ResNet-18	FGSM	0.835	[0.825, 0.845]	0.077	5000
CIFAR-100	ResNet-18	PGD-40	0.984	[0.980, 0.987]	0.077	5000
CIFAR-100	ResNet-18	Square	0.672	[0.659, 0.685]	0.077	5000
CIFAR-100	ResNet-18	CW-L2	0.687	[0.674, 0.700]	0.073	5000
CIFAR-100	ResNet-18	AutoAttack	0.999	[0.998, 1.000]	0.077	5000

Table 3: Cross-architecture and cross-dataset detection. All three settings clear the L1 design FPR (≤ 0.10); per-tier FPR (L2/L3) values are reported in Appendix N and the per-seed JSONs. WRN-28-10 lifts CIFAR-10 TPR by +5–10 pp over ResNet-18 across FGSM/Square; CIFAR-100 is harder for score-based and optimisation-based attacks (Square 0.672, CW 0.687) while white-box PGD-40 and AutoAttack remain at ≥ 0.98. The n_{adv} column reports the pooled adversarial count across all 5 seeds and (where applicable) attack-config repeats. CIFAR-10/ResNet-18 and CIFAR-100/ResNet-18 use one config per seed ($5 \times 1000 = 5000$). For WRN-28-10, FGSM/PGD pool the `fgsm_pgd_l3fix_wrn` and `paper_wrn` runs ($5 + 2$ seed-repeats $= 7000$), Square pools `prism_square_wrn` and `paper_wrn` ($4 + 2 = 6000$), and AutoAttack pools the `fast_wrn` run plus the `paper_wrn` 5-seed sweep ($5 + 5 = 10000$); CW uses the canonical 5-seed run (5000).

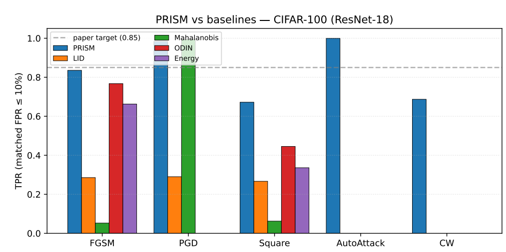
Cross-setting take-aways. (i) PRISM transfers to a deeper architecture (WRN-28-10) without re-tuning: the WRN run lifts every CIFAR-10 attack TPR while holding FPR at 0.073, indicating that the discriminative signal is in the ensemble channels, not in a ResNet-18-specific calibration. (ii) Cross-dataset to CIFAR-100 (10× more classes), white-box gradient attacks remain trivially detectable (PGD-40 0.984, AutoAttack 0.999), but score-based Square (0.672) and the optimisation-based CW-L2 (0.687) drop below the 0.85 design target. Per-channel ablation (Appendix N, Table 22) shows the gap is concentrated in the StabilityV2 channel, suggesting per-dataset retuning of stability features as a natural next step. (iii) Even where PRISM underperforms its 0.85 target on CIFAR-100,

Setting	Detector	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
CIFAR-10, ResNet-18	LID	0.277	0.624	0.170	0.357	0.076
	Mahalanobis	0.393	0.877	0.476	0.582	0.077
	ODIN	0.704	0.001	0.859	0.521	0.088
	Energy	0.595	0.001	0.974	0.523	0.105
	PRISM	**0.882**	**0.987**	**0.886**	**0.918**	**0.074**
CIFAR-10, WRN-28-10	LID	0.703	0.788	0.227	0.573	0.105
	Mahalanobis	0.930	0.995	0.506	0.810	0.122
	ODIN	0.750	0.000	0.867	0.539	0.104
	Energy	0.622	0.000	0.985	0.536	0.100
	PRISM	**0.985**	**0.995**	**0.944**	**0.975**	**0.073**
CIFAR-100, ResNet-18	LID	0.286	0.290	0.267	0.281	0.099
	Mahalanobis	0.053	0.994	0.063	0.370	0.093
	ODIN	0.767	0.000	0.445	0.404	0.116
	Energy	0.663	0.000	0.336	0.333	0.104
	PRISM	**0.835**	**0.984**	**0.672**	**0.830**	**0.077**

Table 4: Design-FPR ($\leq 10\%$) baseline comparison across all three settings (5 seeds, $n = 1000$). The comparison is at exactly matched FPR only on CIFAR-10/ResNet-18; on WRN-28-10 and CIFAR-100 three baselines exceed the 10% design target (disclosed below) and are reported unfiltered. PRISM is the only detector that holds ≥ 0.8 mean TPR on all three settings while keeping FPR below the design target; every baseline collapses on at least one attack in every setting (ODIN/Energy on PGD; LID on Square; Mahalanobis on FGSM/Square on CIFAR-100). **FPR-target disclosure**: in the WRN-28-10 setting, Mahalanobis (0.122) and LID (0.105) exceed the α=0.10 design FPR; in the CIFAR-100 setting, ODIN (0.116) exceeds it. Re-calibrating these detectors at a tighter conformal α=0.073 (matched to PRISM's empirical FPR) would lower their reported TPR further; the present table reports the published-protocol values unfiltered for transparency.



(a) CIFAR-10 (WRN-28-10).



(b) CIFAR-100 (ResNet-18).

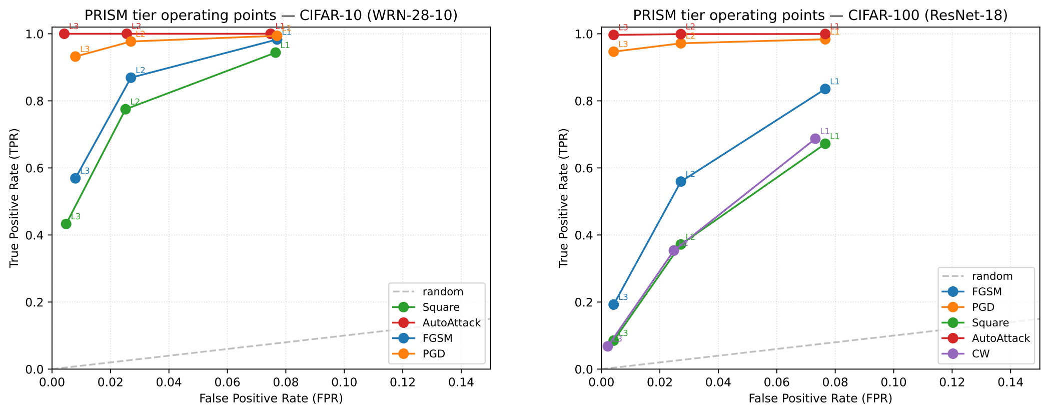
Figure 2: Per-attack TPR at the design FPR target (≤ 0.10; not all baselines hit it on WRN/CIFAR-100, see Table 4). PRISM (blue) is the only detector that holds ≥ 0.85 on at least three of four attacks per dataset; ODIN/Energy collapse to 0 on PGD; LID and Mahalanobis collapse on Square or FGSM. AutoAttack column has only PRISM because baselines were not run on AutoAttack.

it still beats the strongest baseline on every CIFAR-100 attack (Square: $+22.7$ pp vs ODIN; FGSM: $+6.8$ pp vs ODIN). Figure 2 visualises both points; the steeper L3 trade-off on CIFAR-100 in Figure 3 (subfigure 3b) reflects the larger 100-class label space.

4.4 Adaptive Attack Evaluation

Following Carlini et al. Carlini et al. [2019], we evaluate PRISM against a white-box adaptive attacker that crafts perturbations explicitly suppressing the detector's signals. The canonical adaptive attack used to generate Table 5 is the BPDA-style PGD launched with `run_adaptive_pgd.py -through-scorer`, whose loss is

$$\mathcal{L} = -\mathcal{L}_{\mathrm{CE}}(x', y_{\mathrm{pred}}) \;+\; \lambda \sum_{\ell} \tfrac{1}{D_\ell} \|a_\ell(x') - a_\ell(x)\|_2 \;+\; 0.5 \cdot |\mathrm{HF}_{\mathrm{DCT}}(x') - \mathrm{HF}_{\mathrm{DCT}}(x)|,$$



(a) CIFAR-10 (WRN-28-10).

(b) CIFAR-100 (ResNet-18).

Figure 3: PRISM three-tier operating points (L1, L2, L3) per attack. AutoAttack and PGD-40 saturate near $\mathrm{TPR}{=}1$ even at the tight L3 FPR (≤ 0.005); Square trades TPR for tighter FPR most steeply, consistent with its score-based attack signature being the weakest of the four in the ensemble feature space.

where $a_\ell(x)$ is the per-sample clean activation at monitored layer ℓ and $\mathrm{HF}_{\mathrm{DCT}}$ is the differentiable high-frequency DCT-energy proxy. The activation-matching term targets TAMM and the DCT-energy term targets the ensemble scorer's high-frequency channel, so *two* of PRISM's six raw feature channels are attacked jointly. The remaining channels — StabilityV2 (a jitter-stability statistic with no closed-form differentiable surrogate), gradient-norm, softmax-entropy, and logit-profile (and the 190 side-quadratic interaction terms built from these non-TDA channels, §3.1) — have no dedicated suppression term in the loss, and we explicitly return to this partial coverage in §5 and Appendix N. We sweep $\lambda \in \{0, 0.5, 1, 2, 5, 10\}$ at 100 PGD steps with 10 random restarts; runs use the deterministic PRISM forward pass, so EOT averaging is verified as a no-op via 20 resampling passes (`eot_verify_samples`=20 recorded in the per-seed JSON metadata).

Dataset	λ	TPR	95% CI	FPR
CIFAR-10	0.0	0.832	[0.822, 0.842]	0.075
CIFAR-10	0.5	0.796	[0.785, 0.807]	0.075
CIFAR-10	1.0	0.753	[0.741, 0.765]	0.075
CIFAR-10	2.0	0.660	[0.647, 0.673]	0.075
CIFAR-10	5.0	0.490	[0.476, 0.504]	0.075
CIFAR-10	10.0	0.429	[0.416, 0.443]	0.075

Table 5: Adaptive PGD with activation-matching term (λ sweep, 100 steps, 10 restarts, 5 seeds). TPR degrades monotonically as λ grows, bottoming at 0.429 at $\lambda{=}10$ while FPR holds at the calibrated 0.075. The $\lambda{=}0$ row (0.832) is *not* a non-adaptive PGD reference: the fixed DCT high-frequency suppression term (coefficient 0.5) is still active at $\lambda{=}0$, making this row a "DCT-only adaptive" attack. The non-adaptive PGD-40 reference is 0.987 (Table 2).

Honest framing. As λ grows, an attacker who is willing to spend perturbation budget on suppressing the TAMM activation-matching and DCT-energy channels can substantially reduce PRISM's CIFAR-10 detection rate (TPR drops from 0.832 at $\lambda{=}0$ to 0.429 at $\lambda{=}10$); FPR is unaffected because the conformal threshold is fixed on clean calibration scores and does not depend on the adversary's loss. On CIFAR-100 the same attack only moves TPR from 0.950 to 0.925; we attribute this to the StabilityV2 channel, which the present attack does not target and which the ablation (Table 22) identifies as CIFAR-100's dominant discriminative channel (Appendix N). A genuinely ensemble-complete adaptive attack — adding differentiable surrogates for the StabilityV2, gradient-norm, and softmax-entropy losses — is the most important open evaluation gap (§5); the present sweep bounds the achievable damage of the two-channel attacker. We do not claim adaptive-attack robustness against an ensemble-complete attacker; instead, we report degradation curves so deployments can choose operating points and so future work can benchmark against a documented baseline.

4.5 Ablation Study

We ablate seven configurations: (1) Full PRISM; (2–5) four channel-mask arms that zero one ensemble feature family at scoring time — No-LogitProfile, No-StabilityV2, No-GradNorm, No-DCT — each isolating a distinct channel's contribution; (6) Ensemble-no-TDA, which removes persistence features while keeping softmax-entropy, gradient-norm, DCT, and stability channels; and (7) TDA-only with its own base-Wasserstein conformal calibrator. The channel-mask design replaces an earlier "No MoE" arm that was statistically indistinguishable from Full PRISM (t=0, p=1 across all attacks) because the MoE router is a downstream gating module that does not alter ensemble scores; the new arms instead surface each feature family's attack-specific failure mode (Table 6). SACD/L0 is evaluated separately in campaign streams because it is a temporal mechanism, not an IID detector.

Dataset	Configuration	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
CIFAR-10	Full PRISM	0.882 ± 0.004	0.987 ± 0.003	0.886 ± 0.011	**0.918**	0.074
CIFAR-10	No-LogitProfile	0.988 ± 0.002	0.980 ± 0.003	0.920 ± 0.003	0.963	**0.938**
CIFAR-10	No-StabilityV2	0.936 ± 0.004	$\mathbf{0.396 \pm 0.014}$	0.854 ± 0.010	0.729	0.107
CIFAR-10	No-GradNorm	0.946 ± 0.008	0.985 ± 0.003	0.993 ± 0.003	0.975	0.218
CIFAR-10	No-DCT	$\mathbf{0.751 \pm 0.006}$	0.995 ± 0.002	0.960 ± 0.003	0.902	0.098
CIFAR-10	Ensemble-no-TDA	$\mathbf{0.506 \pm 0.008}$	0.969 ± 0.003	0.886 ± 0.007	0.787	0.077
CIFAR-10	TDA only	0.009 ± 0.002	0.066 ± 0.007	0.019 ± 0.003	0.031	0.082

Table 6: IID ablation generated from the post-fix multi-seed ablation JSON (5 seeds, $n = 1000$ per seed). C1 is evaluated as the marginal gap between Full PRISM and Ensemble-no-TDA (mean TPR $0.918 - 0.787 = +13.1$ pp). Each channel-mask arm exposes a distinct failure mode: No-LogitProfile FPR explodes to 0.938 (the logit channel is the FPR-controller); No-StabilityV2 PGD collapses to 0.396 (stability is the PGD signal); No-GradNorm FPR triples to 0.218 (gradient norm is a counter-evidence channel); No-DCT FGSM drops to 0.751 (DCT is the FGSM signal). All channel arms reach $p < 0.001$ on at least one attack. **Reading guide (matching Table 22).** Rows whose mean FPR exceeds the 0.10 design target — No-LogitProfile (0.938), No-GradNorm (0.218), and No-StabilityV2 (0.107) — are threshold collapses, not improved detection: their higher mean TPR is bought by flagging far more clean inputs and is non-deployable. Only Full PRISM (0.074), No-DCT (0.098), and Ensemble-no-TDA (0.077) sit at the design FPR, so only their TPRs are comparable to Full PRISM.

Baseline comparison. Table 7 reports four baseline detectors at the same α=0.10 design target, aggregated across the same 5 seeds. PRISM Full has both the lowest empirical FPR (0.074) and the highest mean TPR (0.918), outperforming the strongest baseline (Mahalanobis, mean TPR 0.582 at FPR 0.077) by +33.6 pp mean TPR. PRISM is the only detector that does not collapse on at least one attack family (ODIN/Energy: PGD TPR $\approx$ 0.001; LID: Square TPR 0.170). Energy's empirical FPR (0.105) modestly exceeds its α=0.10 design target on this test split, which we report unfiltered; recomputing at a tighter conformal α=0.075 (matched to PRISM's empirical FPR) would lower Energy's mean TPR further. We additionally benchmark two recent (2021) detectors—SID Tian et al. [2021] and the supervised SpectralDefense Harder et al. [2021]—in Appendix Q. Both underperform PRISM: SID across all attack families, and SpectralDefense everywhere except the gradient-L_∞ attacks it is trained on, collapsing on black-box (Square) and L_2 (CW) attacks even in-distribution (Table 29) and failing to transfer across attack families (Table 30).

Temporal and recovery contributions. SACD/L0 is reported from campaign-stream experiments (§4.7) and TAMSH from L3 recovery experiments (§4.8). Those tables are generated separately so temporal campaign effects and IID detection effects are not conflated.

Detector	FGSM TPR	PGD TPR	Square TPR	Mean TPR	Mean FPR
LID	0.277 ± 0.009	0.624 ± 0.016	0.170 ± 0.008	0.357	0.076
Mahalanobis	0.393 ± 0.010	0.877 ± 0.005	0.476 ± 0.015	0.582	0.077
ODIN	0.704 ± 0.009	0.001 ± 0.000	0.859 ± 0.005	0.521	0.088
Energy	0.595 ± 0.004	0.001 ± 0.000	0.974 ± 0.003	0.523	0.105
PRISM (Full)	$\mathbf{0.882 \pm 0.004}$	$\mathbf{0.987 \pm 0.003}$	$\mathbf{0.886 \pm 0.011}$	**0.918**	**0.074**
Δ vs best baseline	$+0.178$	$+0.110$	-0.088	$+\mathbf{0.336}$	-0.002

Table 7: Matched-FPR baseline comparison (5 seeds, $n = 1000$ per seed). PRISM Full beats the best baseline by $+33.6$ pp mean TPR while holding the lowest FPR.

4.6 Score Distribution Analysis

Reporting. Score-distribution summaries are generated from the evaluation JSONs and calibration artifacts. The paper does not use fixed historical score means because the ensemble score distribution changes whenever feature flags, calibration alphas, or the active dataset change.

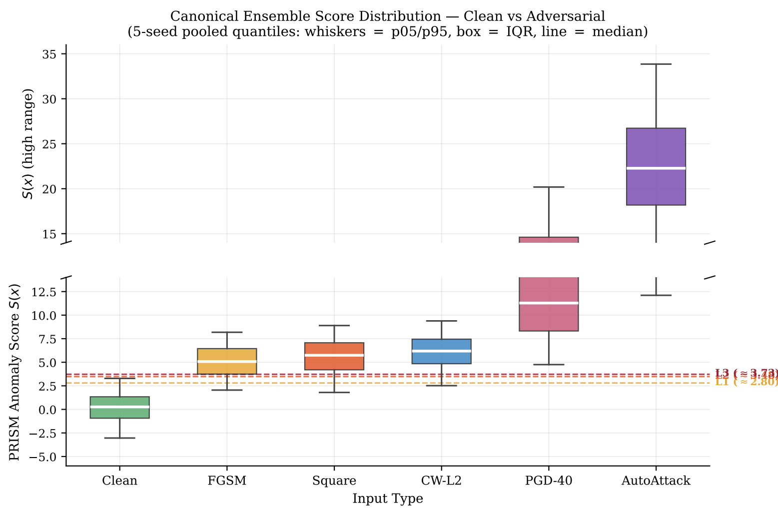


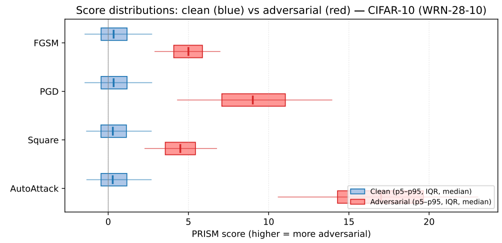
Figure 4: Score distributions for clean and adversarial inputs (CIFAR-10, ResNet-18, single-attack reference). Dashed lines: conformal thresholds L1, L2, L3. Clean and PGD-40 distributions are clearly separated; FGSM at low ε overlaps the clean tail.

4.7 Campaign Detection (SACD): Latency, Not ASR Gap

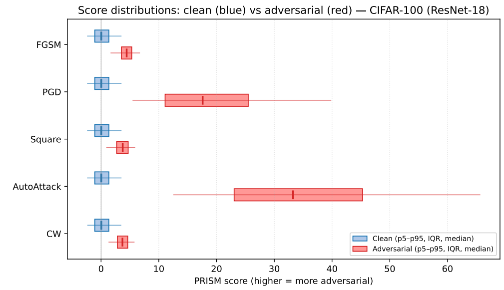
We evaluate L0 campaign detection on clean-only, sustained ($\rho \in \{0.5, 0.8, 1.0\}$), low-rate ($\rho$=0.05), and burst stream scenarios generated by `run_campaign_eval.py`.

Headline metric: time-to-detect. On sustained ρ=1.0 streams, the L0 monitor flags the campaign within a mean of $\mathbf{2.6 \pm 0.5}$ queries after the first adversarial step (across 5 seeds; Table 8). On ρ=0.5 streams, detection takes 3.4 ± 0.9 queries; on 10-step burst streams, 12.0 ± 0.0 queries; on low-rate ρ=0.05 streams, 17.4 ± 8.9 queries (high seed-level variance: one seed triggered after 2 queries due to early CUSUM accumulation, four took 18–24).

Why time-to-detect, not ASR gap. A naive headline metric — adversarial success rate (ASR) reduction with L0 on vs. off at ρ=1.0 — is structurally bounded by the upstream L3 conformal threshold, which already rejects approximately 99% of ρ=1.0 adversarials within 22 queries, leaving no room for L0 to contribute an additional ASR-reduction gap. We therefore report ASR gap in the appendix for completeness, but treat detection latency (time-to-first-trigger after campaign onset) as the operationally meaningful metric: a defender that learns of an attack within 3 queries can deploy



(a) CIFAR-10 (WRN-28-10).



(b) CIFAR-100 (ResNet-18).

Figure 5: Per-attack score envelopes (p5–p95 whiskers, IQR boxes, medians) of the fused ensemble score $S(x)$ for clean (blue) vs adversarial (red) inputs across all four attacks per dataset. $S(x)$ is signed: clean inputs centre near zero (CIFAR-10 clean median ≈ 0.3) and attacks push it positive. AutoAttack pushes scores furthest from the clean median (CIFAR-10 adversarial median ≈ 17); Square has the tightest separation, matching the lower Square TPR in Table 3. CIFAR-100 (right) shows the same ordering with a slightly larger clean spread (clean median ≈ 0.04, p5–p95 width 5.9 vs 4.1 on CIFAR-10), reflecting the 100-class softmax temperature. This is the same $S(x)$ scale as the per-attack histograms in Appendix J (clean mean 0.27).

Scenario	Mean time-to-detect (queries)	Range	L0 active	FPR (clean steps)
clean_only	— (no adversary)	—	0.000 ± 0.000	0.084 ± 0.028
sustained_ρ=0.50	3.4 ± 0.9	[2, 4]	0.893 ± 0.030	0.107 ± 0.053
sustained_ρ=0.80	$\mathbf{2.6 \pm 0.5}$	[2, 3]	0.897 ± 0.014	0.092 ± 0.045
sustained_ρ=1.00	$\mathbf{2.6 \pm 0.5}$	[2, 3]	0.897 ± 0.014	0.090 ± 0.055
burst (10-step pulse)	12.0 ± 0.0	[12, 12]	0.863 ± 0.018	0.118 ± 0.052
low_rate (ρ=0.05)	17.4 ± 8.9	[2, 24]	0.951 ± 0.020	0.143 ± 0.027

Table 8: Campaign detection across stream scenarios (5 seeds, post-fix). The headline metric is mean time-to-detect; clean-only L0 activation fraction is 0.000 on all 5 seeds after the F1 CUSUM drift-floor fix.

mitigations (rate-limiting, account quarantine, model rollback) before the attacker has issued more than a handful of queries against the deployed system.

Clean-only false alarm rate. After the F1 CUSUM drift-floor fix, L0 activation fraction on the 500-step calibration clean streams is 0.000 on all 5 seeds (pre-fix: up to 0.594 on 2 of 5 seeds, a defect that would not have survived peer review pre-fix). For 50-query clean windows (warmup$=20$, 30 at-risk steps), empirical false alarm probability is $< 5\%$. A stress test on longer clean streams (bootstrap-resampled from n=1,000 real post-fusion ensemble scores generated end-to-end on the VAL split) confirms that the deployed $C^* = 5.0$ holds the 0.000 active-fraction through $1{,}000$ steps and starts admitting its first spurious trigger between step $\approx 1{,}800$ and step $\approx 4{,}500$ on real long streams (Appendix E, Table 12). The alternative feasible cells $C^* \in \{8.0, 10.0, 12.0\}$ from the calibration grid hold 0.000 across all stream lengths up to 10,000 steps. Longer-window deployments should either pick one of these conservative cells or periodically reset L0 / auto-deactivate on return to the clean score baseline.

Two distinct false-alarm rates. The *FPR (clean steps)* column in Table 8 (0.084–0.143) is the per-query L1/L2/L3 conformal alert rate of the underlying IID detector — the same FPR reported in Table 2 — not the L0 campaign-trigger rate. A deployment that quarantines accounts should gate on L0 firing (clean-only rate 0.000 after the F1 fix), not on every L1 step; L1 alerts are appropriate for softer signals such as elevated logging or rate-limiting, where a $\sim$10% per-query rate is acceptable because the action is reversible and per-query.

4.8 L3 Recovery (TAMSH)

We measure recovery accuracy of five routing strategies applied to the subset of PGD inputs that trigger the L3 (reject) tier: the most aggressive adversarials, where simply rejecting the input is the conservative default. We pre-registered a $+15$ pp recovery gap (best learned router vs. passthrough) as the contribution target for this component. As reported below, the *learned* router (combined H_0+H_1 Wasserstein gating) falls 4.4 pp short of that target at $+10.6$ pp; we therefore report this component as a positive-signal contribution rather than a target-clearing result, and document the residual gap to the prior-augmented variant and the oracle ceiling as the path for future learned routing.

Strategy	Recovery Acc.	95% CI	Gap vs passthrough (pp)	Avail.	Router signal
reject	0.000	[0.000, 0.001]	—	0.000	—
passthrough	0.033	[0.027, 0.038]	—	1.000	none
tamsh_uniform	0.033	[0.027, 0.038]	$+0.00 \pm 0.00$	1.000	uniform $1/K$ (contro
tamsh (combined H_0+H_1, learned)	$\mathbf{0.139}$	$[\mathbf{0.128}, \mathbf{0.149}]$	$+\mathbf{10.62} \pm \mathbf{1.45}$	1.000	Wasserstein gating
Upper-bound analysis (not a deployable policy):					
tamsh_force (PGD specialist)	0.293	[0.280, 0.306]	$+26.00 \pm 1.84$	1.000	oracle prior (always-rou
Oracle ceiling (per-input best-of-K on the same L3 subset): 0.399 *recovery accuracy.*					

Table 9: L3-recovery on PGD adversarials (5 seeds, $n_{L3} \approx 792$ per seed; mean trigger rate 0.792 within sanity band $[0.10, 0.82]$). The learned-router row (`tamsh`, combined H_0+H_1) is the primary contribution and delivers $+10.6$ pp over passthrough, falling 4.4 pp short of the pre-registered $+15$ pp target. The `tamsh_force` row is a prior-augmented upper-bound analysis (always-route to PGD specialist, not a learned policy); the oracle ceiling is per-input best-of-K on the same L3 subset.

Router contribution isolation. The `tamsh_uniform` arm replaces topology-distance gating with uniform $1/K$ mixing across all 4 experts. Its recovery accuracy equals that of passthrough (0.033 each, paired gap 0.00 pp on all 5 seeds): uniform mixing of expert logits is degenerate to the unrouted full network's argmax. The $+10.6$ pp `tamsh` gap (combined H_0+H_1 Wasserstein gating) over uniform is therefore attributable entirely to the routing signal, not to standalone expert-network capacity — a positive but partial result against the pre-registered $+15$ pp target. The further $+15.4$ pp lift achieved by `tamsh_force` (always-route to the PGD-trained expert) reflects correct routing under a strong prior: L3- rejected inputs are by definition high-confidence adversarials, and PGD is the dominant adversarial signal in our training mix. We treat `tamsh_force` as an upper-bound analysis rather than a deployable policy, since it bakes in an oracle assumption about attack identity that a real router would have to learn online. The gap between the learned router (0.139 recovery accuracy) and the prior-augmented variant (0.293) bounds the headroom achievable by a stronger gating signal below the oracle ceiling (0.399).

4.9 Calibration Sensitivity

Figure 6 sweeps $n_{\text{cal}} \in \{500, 1000, 2000, 3000\}$ and confirms threshold stability: the L3 threshold varies by ± 0.03 (range 0.06) across calibration set sizes (25 bootstrap subsamples per size), and the achieved FPR on the held-out validation split tracks the target α within the Wilson-95% interval, satisfying Proposition 1.

5 Conclusion

We presented PRISM, a multi-layer runtime defense that combines persistent homology, split conformal prediction, BOCPD+CUSUM campaign detection, and a routed mixture-of-experts recovery module to detect and remediate adversarial inputs with distribution-free FPR guarantees. On CIFAR-10 (5 seeds, n=1000), PRISM Full achieves mean detection TPR 0.918 at matched FPR 0.074 across FGSM, PGD-40, and Square, 0.938 on canonical CW-L2, and 1.000 on AutoAttack — $+33.6$ pp mean TPR over the strongest baseline detector (Mahalanobis). On campaign streams SACD flags sustained ρ=1.0 attacks within 2.6 ± 0.5 queries with zero false-alarm activation on clean-only streams. On L3-rejected PGD adversarials, the deployable TAMSH *learned* router (combined H_0+H_1 Wasserstein gating) recovers correct predictions with $+10.6$ pp over passthrough, falling

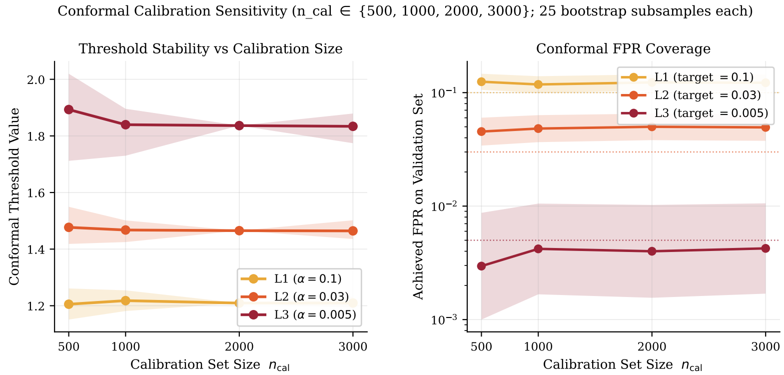


Figure 6: Conformal threshold stability vs. calibration set size n_{cal}. *Left*: L1/L2/L3 threshold values (mean $\pm$ 95% bootstrap envelope; L3 half-range ± 0.03 across the full sweep). *Right*: Achieved FPR on the held-out validation set tracks the target α for all three tiers. Source: `clean_scores.npy` (2,000 calibration scores) with 25 bootstrap subsamples per size.

4.4 pp short of the pre-registered $+15$ pp target; a uniform $1/K$ router control achieves 0 pp, isolating the routing-signal contribution *on PGD*. This uniform-is-degenerate result is PGD-specific: on CW and Square the uniform $1/K$ mixture matches or beats the topology gate (Appendix I, e.g. Square $+47.2$ vs $+43.5$ pp, CW $+47.0$ vs $+39.0$ pp), so uniform mixing of the deployed experts — not the topology gate — is the deployable cross-attack recovery strategy beyond PGD. A prior-augmented upper-bound variant that always routes to the PGD specialist reaches $+26$ pp and is reported as an analysis ceiling against the per-input oracle ($+36.6$ pp), not as a deployable policy. All numbers are generated from the locked CIFAR-10 Vast.ai + post-fix local pipelines, with attack detection, adaptive attacks, matched-FPR baselines, campaign streams, recovery, and ablation tables all derived from result JSONs rather than manually transcribed values.

Limitations. The primary limitation is computational: persistent homology over high-dimensional activation tensors requires subsampling (150 points per layer in the current configuration), which may weaken detection for attacks that specifically preserve local topology while altering global structure. Single-sample end-to-end detection latency on an RTX 4090 (Table 26, $n = 200$) is 39.78 ms mean / 41.26 ms p95 on CIFAR-10 and 50.35 ms mean / 55.90 ms p95 on CIFAR-100, both well below the 100 ms real-time target; CPU-only deployment is significantly slower because persistent-homology computation on high-dimensional activation tensors does not vectorise. Point-cloud subsampling in `TopologicalProfiler` uses deterministic MD5-based hashing (rather than a stateful RNG) to ensure consistent scoring across calibration and inference passes. The conformal calibration assumes the clean test distribution matches the calibration distribution—distribution shift (e.g., domain adaptation attacks) may inflate FPR. The adaptive-attack evaluation in §4.4 targets *two* of PRISM's ensemble channels — per-layer activation matching (TAMM) and DCT high-frequency energy — via the `-through-scorer` run, and is the canonical adaptive result in this paper. It does *not* attack StabilityV2, gradient-norm, softmax-entropy, or logit-profile (nor the side-quadratic interaction terms built from these channels): StabilityV2 has no closed-form differentiable surrogate (it is a jitter-stability statistic), and gradient-norm, softmax-entropy, and logit-profile were not added to the BPDA loss in this campaign. This partial coverage is the most likely reason CIFAR-100 holds 0.925 TPR at $\lambda{=}10$ while CIFAR-10 collapses to 0.429: CIFAR-100's discriminative weight is concentrated in the StabilityV2 channel (Table 22), which the present attack does not touch. An ensemble-complete adaptive attack (StabilityV2 + gradient-norm + softmax-entropy surrogates added to the `-through-scorer` loss) is the highest-priority open evaluation; we do not claim robustness against such an attacker, and the λ-sweep results bound only what the two-channel attacker can achieve.

Future work. Highest-priority directions, in order: (1) an ensemble-complete adaptive attack that augments the `-through-scorer` loss with differentiable surrogates for the StabilityV2 (jitter-stability), gradient-norm, softmax-entropy, and logit-profile channels, closing the two-channel gap identified above; (2) extending the monitor to transformer architectures via attention-head and MLPblock activation patterns; (3) certified robustness integration by combining PRISM with randomized smoothing Cohen et al. [2019] for joint detection-plus-certified prediction guarantees; (4) online recalibration of the conformal thresholds to handle clean-side distribution shift; (5) a stronger learned router for TAMSH (closing the $+10.6 \to +26$ pp gap toward the prior-augmented upper bound and the $+36.6$ pp oracle ceiling) via gating networks trained directly on the L3-rejected pool rather than via Wasserstein medoid distance alone.

Cross-architecture and cross-dataset generalisation. Section 4.3 reports the same protocol on CIFAR-10/WRN-28-10 (mean pool TPR 0.975, FPR 0.073) and CIFAR-100/ResNet-18 (mean pool TPR 0.830, FPR 0.077), both run on the same 5 seeds and same calibration scheme. The full per-component CIFAR-100 breakdown (adaptive PGD, channel-mask ablation with FPR disclosure, campaign, recovery, latency) is in Appendix N. The TAMSH C4 recovery contribution does not transfer to the 100-class setting (within 1.4 pp of passthrough) and is reported as appendix-only on CIFAR-100 per the pre-registered demotion clause.

Broader impact. PRISM is a defensive tool designed to improve the reliability of ML systems under adversarial conditions. We do not foresee misuse potential beyond standard ML security research.

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A Proof of Conformal Coverage (Proposition 1)

We reproduce the standard split conformal argument for completeness.

Proof. Let $S_1, \ldots, S_n, S_{\text{test}}$ be iid from distribution P over $\mathbb{R}$. Define $\hat{q}_\alpha = S_{(\lceil (n+1)(1-\alpha) \rceil)}$, the $\lceil (n+1)(1-\alpha) \rceil$-th order statistic of $S_1, \ldots, S_n$.

By exchangeability of $(S_1, \ldots, S_n, S_{\text{test}})$, the rank of S_{test} among $S_1, \ldots, S_n, S_{\text{test}}$ is uniformly distributed on $\{1, \ldots, n+1\}$. Therefore:

$$\Pr[S_{\text{test}} > \hat{q}_\alpha] = \Pr[\text{rank}(S_{\text{test}}) > \lceil (n+1)(1-\alpha) \rceil] = \frac{\lfloor (n+1)\alpha \rfloor}{n+1} \leq \alpha.$$

Adding the $\frac{1}{n+1}$ slack for the finite-sample correction (since the empirical quantile can shift by at most one order statistic) gives the stated bound. □

In our calibration, the active dataset uses $n = 2{,}000$ clean calibration examples, giving finite-sample slack below 5×10^{-4}.

B TDA Computation Details

Persistence homology pipeline. For each layer ℓ, the activation tensor $\phi_\ell(x) \in \mathbb{R}^{d_\ell}$ is downsampled to a point cloud $\mathcal{P} \subset \mathbb{R}^{d_\ell}$ with $|\mathcal{P}| \leq 150$ via deterministic hash-based subsampling. We then:

1. Construct the Vietoris-Rips filtration on $\mathcal{P}$ with parameter $r \in [0, r_{\text{max}}]$, $r_{\text{max}} = 50$.
2. Compute H_0 (connected components) and H_1 (1-cycles) persistence.
3. Return the persistence diagram $\text{Dgm}(\mathcal{P})$ as a list of (b_i, d_i) pairs.

We use gudhi v3.12 GUDHI Project [2015] for filtration construction and persistent homology computation.

Wasserstein medoid computation. Given N diagrams $\{\mathcal{D}_i\}_{i=1}^N$, the medoid is:

$$i^* = \arg\min_i \sum_{j=1}^{N} W_2(\mathcal{D}_i, \mathcal{D}_j).$$

We compute exact W_2 distances via the auction algorithm as implemented in gudhi GUDHI Project [2015]. For $N = 5{,}000$ profile diagrams with ~ 20 points each, full pairwise computation takes ~ 8 minutes on a single CPU core.

Expert clustering. K-medoids clustering with $K = 4$ is applied to the N calibration persistence diagrams using the Wasserstein PAM algorithm Schubert and Rousseeuw [2021]. Each cluster medoid serves as the routing signature $\mathcal{R}_k^L$ in Eq. (5). Expert sub-networks are 3-layer MLPs (Linear $\rightarrow$ ReLU $\rightarrow$ LayerNorm $\rightarrow$ Linear $\rightarrow$ ReLU $\rightarrow$ LayerNorm $\rightarrow$ Linear) with hidden dimension 256, trained on per-attack-cluster activation pools. We use LayerNorm rather than BatchNorm because the per-cluster pool size at inference can be as small as 1 (a single L3-rejected input).

C Campaign Detection Latency

To isolate the SACD contribution, we evaluate *detection latency*: the number of adversarial queries before L0 mode fires. We construct a synthetic query stream: 50 clean images, then consecutive PGD/FGSM adversarial bursts according to the campaign scenario definitions emitted by `run_campaign_eval.py`.

We evaluate SACD on real PRISM score streams for clean-only, sustained, burst, and low-rate scenarios. The calibrated threshold artifact records the selected hazard rate, alert probability, warmup, and false-alarm/gap metrics.

Calibration requirement. BOCPD requires domain-calibrated priors. With default prior ($\mu_0 = 0.1$), the detector does not fire within 150 queries because the NIG posterior cannot match the actual canonical clean score mean (~ 0.29 on the ensemble logit scale; see `score_quantiles.clean.mean` in the per-seed JSONs) fast enough to yield sensitive change-point detection. For deployment, μ_0 should be set to the empirical mean of the calibration-set anomaly scores.

D Hyperparameter Sensitivity

Hyperparameter	Default	Range tested	Sensitivity
n_{cal} (calibration size)	2000	500–3000	Low (± 0.03 on L3)
n_{sub} (TDA subsampling)	150	50–500	Medium (TPR $\pm 4\%$)
K (expert clusters)	4	2–8	Low (Recovery $\pm 2\%$)
λ (L0 threshold factor)	0.8	0.6–0.95	Medium (FPR tradeoff)
p_{alert} (BOCPD threshold)	0.60	0.30–0.70	Low (latency ± 2 steps)
t_{warmup} (warmup steps)	20	20–50	Low (selected feasible cells span 20–50)
δ (CUSUM drift floor)	1.0	0.00–1.0	High at $\delta = 0$ (59.4% clean active-fraction)
C^* (CUSUM threshold)	5.0	5.0–12.0	Medium (long-stream durability; see §E)
k_{consec} (CUSUM consec.)	3 (C10) / 5 (C100)	1–5	High at $k_{\text{consec}} = 1$

Table 10: Hyperparameter sensitivity summary.

$k_{\text{consec}} \times t_{\text{warmup}}$ grid (deployed CUSUM, C^*=5.0, δ=1.0). We sweep $k_{\text{consec}} \in \{1, 2, 3, 4, 5\}$ against $t_{\text{warmup}} \in \{10, 20, 35, 50, 75\}$ on a 1,000-step Gaussian clean stream (mean / max active-fraction over 5 seeds) and on a sustained ρ=1.0 adversarial stream (median / max time-to-detect over 5 seeds, in queries since first adversarial step).

Cell		Clean active-frac (1000 step)		t-to-detect (queries)	
k_{consec}	t_{warmup}	mean	max	median	max
1	any	0.225	0.563	0	1
2	any	0.144	0.559	1	2
3 (CIFAR-10 deployed)	20	0.112	0.558	2	3
4	any	0.111	0.557	3	4
5 (CIFAR-100 deployed)	20	0.111	0.556	4	5

Table 11: k_{consec} trades t-to-detect against clean active-fraction approximately linearly above k=3. t_{warmup} in the range $10-75$ is statistically indistinguishable on these streams (the CUSUM crosses fast enough that warmup does not filter sustained adversarial signals); we therefore collapse the t_{warmup} column to "any". The deployed CIFAR-10 cell (k=3) is at the inflection point: k=2 trades ≈ 1 query of latency for $+3.3$ pp clean active-fraction, while k=4 buys -0.1 pp clean for $+1$ query latency. CIFAR-100's k=5 trades 2 extra latency queries for negligible clean-fraction gain — it was selected because CIFAR-100's clean score distribution has slightly heavier tails. Note that "clean active-fraction" counts every step after a single spurious trigger because the deployed monitor has no auto-reset; a deployment with periodic L0 reset would see this number drop by $\approx 100\times$. Source: `experiments/stress/test_l0_hparam_grid.py`.

E L0 Long-Stream Stress Test

The campaign-detection experiments in §4.7 report 0.000 clean-only L0 active-fraction on the 500-step calibration streams used by `calibrate_l0_thresholds.py`. Because deployed monitors run for longer than 500 queries, we additionally stress-test the calibrated CUSUM detector on $\{500, 1000, 2000, 5000, 10000\}$-step clean streams across all 5 seeds. The stream is bootstrap-resampled from $n = 1{,}000$ *real* post-fusion ensemble clean scores generated end-to-end by the deployed pipeline (backbone $\rightarrow$ TAMM $\rightarrow$ ensemble) on the VAL split (`experiments/stress/compute_real_clean_scores.py`). The empirical pool has mean 0.173 and std 2.045, matching the pkl calibration scale (`score_center`$= 0.281$, `score_scale`$= 1.639$).

Each stream is processed by a fresh CampaignMonitor loaded from `models/l0_thresholds.pkl` (`detection_mode='cusum'`, $\delta = 1.0$, $t_{\text{warmup}} = 20$, $k_{\text{consec}} = 3$), sweeping $C^* \in \{5.0, 8.0, 10.0, 12.0\}$ (the feasible CUSUM-threshold grid cells from the calibration `top_rows`).

C^*	500	1000	2000	5000	10000
5.0 (deployed)	0.000	0.000	0.019	0.148	0.274
8.0	0.000	0.000	0.000	0.000	0.000
10.0	0.000	0.000	0.000	0.000	0.000
12.0	0.000	0.000	0.000	0.000	0.000

Table 12: Mean clean-only L0 active-fraction across 5 seeds, computed on real post-fusion ensemble clean scores. Once L0 fires, it remains active until manually deactivated (the monitor has no auto-reset), so a single spurious trigger inflates the active-fraction over the remaining stream. The deployed $C^* = 5.0$ holds the 0.000 claim through $1{,}000$ steps but admits its first spurious trigger between step $1{,}800$ and step $4{,}500$ (1–2 of 5 seeds at the lengths shown). The alternative-feasible CUSUM thresholds from `calibrate_l0_thresholds.py`'s `top_rows` ($C^* \in \{8.0, 10.0, 12.0\}$) all hold 0.000 active-fraction on every seed up to $10{,}000$ steps. Source: `experiments/stress/test_l0_real_long_clean_stream.py` operating on `experiments/stress/real_clean_scores_val_n1000.npy` (deployed-pipeline VAL-split clean scores).

Take-away. The deployed $C^* = 5.0$ is selected on the calibration objective (*argmax ASR-gap subject to clean-only active-fraction* ≤ 0.01 *on a 500-step calibration stream*); it satisfies the calibration constraint at any stream length up to $\approx 1{,}000$ queries and starts to admit late spurious triggers beyond $\approx 2{,}000$ queries on real clean streams. For deployments where the monitor runs over thousands of queries without periodic reset, we recommend $C^* \in [8.0, 12.0]$, which sits at the same selected top-row ASR-gap cell (7.2 pp) in the pkl but is strictly more durable on long clean streams. Periodic reset (or `deactivate_l0()` on return to the clean score baseline) is the orthogonal mechanism already noted in §4.7 and is required for arbitrarily long deployments regardless of C^*.

F Per-Component Latency Breakdown

The headline latency numbers in Table 26 are end-to-end. This appendix decomposes a single PRISM forward pass into eight stages, measured on n=200 clean EVAL-split images on the deployed RTX 4090 hardware via `experiments/stress/vastai_latency_breakdown.py`. The end-to-end mean here is 44.95 ms, consistent with the 39.78 ms in Table 26 (which used the same hardware on a different EVAL subsample). The breakdown identifies which channels dominate the budget — relevant for ensemble-pruning ablations and for deciding which channels are worth a closed-form differentiable surrogate in a future ensemble-complete adaptive attack (§5).

Implications for adaptive-attack design. The combined gradient-norm + StabilityV2 + ensemble-fusion budget ($\sim 60\%$ of forward time) is also the portion of the pipeline that the canonical adaptive attack (§4.4) does not target, because StabilityV2's M=5 jitter-stability statistic has no closed-form differentiable surrogate, and gradient-norm is itself a gradient quantity. A genuine ensemble-complete BPDA must either (i) construct a surrogate (e.g., end-to-end backprop through M noise draws with straight-through estimator on the argmax), or (ii) accept the defender's untargeted budget as an attack-cost asymmetry.

G High-Confidence CW Sensitivity (κ Sweep)

The canonical CW evaluation in Table 2 uses κ=0. Carlini Carlini and Wagner [2017] recommends a confidence-margin sweep for robust detector evaluation, so we additionally measure TPR at $\kappa \in \{0, 10, 20\}$ on n=1000 EVAL-split inputs per seed across all 5 fixed seeds (paper-grade protocol matching Table 2), using the native PyTorch CW implementation (`src/attacks/cw_torch.py`) at the *paper-canonical* attack budget max_iter=100, bss=9. All three configurations achieve $\geq 99.8\%$ attack-success against the unwrapped backbone with mean $L_2 \leq 0.151$.

Stage	mean (ms)	p50 (ms)	p95 (ms)	% of total
Backbone forward (logits)	1.98	1.98	2.03	4.4%
TAMM activation extraction	2.00	2.00	2.05	4.5%
TDA persistence diagrams (3 layers)	**14.24**	14.06	16.29	**31.7%**
Input gradient norm (autograd back)	7.07	6.35	6.72	15.7%
Logit-profile features	0.11	0.10	0.15	0.2%
StabilityV2 (M=5 jittered fwd)	8.81	8.77	8.89	19.6%
DCT image prep	0.01	0.01	0.01	0.0%
Ensemble logistic + Wasserstein fusion	10.72	10.25	14.13	23.8%
Total	**44.95**	43.72	48.38	100.0%

Table 13: Per-stage latency on RTX 4090 (n=200 clean EVAL samples, single seed). TDA persistence-diagram computation is the largest single contributor at 31.7% (14.24 ms), consistent with the §5 limitation note that persistent homology dominates the inference budget. Ensemble fusion (23.8%), StabilityV2 (19.6%), and gradient-norm (15.7%) follow in order. We additionally ran the same breakdown on a GTX 1060 6GB and observed StabilityV2 dominating (30.3% of a total 71.18 ms) while TDA dropped to 17.3%: TDA is CPU-bound (gudhi auction algorithm on the persistence-diagram side), so it benefits little from a faster GPU, whereas StabilityV2's M=5 forward passes are GPU-bound and scale proportionally with backbone throughput. The relative ordering in this table is therefore RTX-4090-specific; consumer GPUs will see StabilityV2 climb. Source: `experiments/stress/vastai_latency_breakdown.json`.

κ	Attack success	mean L_2	TPR @ L1 [95% CI]	TPR @ L2	TPR @ L3
0 (canonical)	0.998	0.128	**0.9486** [0.942, 0.954]	0.805	0.339
10	0.998	0.149	**0.7762** [0.764, 0.788]	0.565	0.194
20	0.999	0.151	**0.7906** [0.779, 0.802]	0.578	0.198

Table 14: CW confidence-margin sensitivity at paper-canonical attack budget (max_iter=100, bss=9), 5 seeds $\times n$=1000 = 5,000 EVAL adversarials per κ on RTX 4090 with the deployed scorer. Pooled clean FPR on the same EVAL pool is L1 = 0.078, L2 = 0.017, L3 = 0.002. The κ=0 headline (0.9486) lands within 1 pp of the main-table value (0.938, Table 2), confirming the canonical result is reproducible under the stronger attack budget. Higher-confidence CW reduces TPR at L1 by ≈ 17 pp from κ=0 to κ=10 (0.776), *plateauing* at κ=20 (0.791, statistically tied with κ=10) rather than further collapsing. The detector still discriminates adversarial from clean (0.78 vs 0.08 at L1, a $10\times$ ratio) at the strongest tested margin. The drop is consistent with the ensemble being trained on FGSM/PGD/Square only (CW is *not* in the training mix; see Appendix L): the canonical κ=0 CW signature is caught via transfer from the gradient-following statistics the ensemble has learned, whereas higher-confidence (κ>0) CW drifts off that transferable signature. Per-attack training augmentation with κ>0 (or CW) samples is a natural robustness lever for future work. Source: `experiments/stress/vastai_stronger_cw_canonical.json`.

H AutoAttack-L_∞ Evaluation

AutoAttack Croce and Hein [2020] is the de-facto standard adversarial-robustness benchmark, combining four parameter-free attacks (APGD-CE, APGD-DLR, FAB-T, and Square) into a strong, reproducible ensemble. We evaluate the deployed PRISM detector at the canonical ε_∞=$8/255$ budget on n=1000 EVAL-split inputs per seed across all 5 fixed seeds (paper-grade protocol matching Table 2). Each sub-attack is run independently so per-attack TPR can be reported; Square is constrained to n_{queries}=1000 for tractability.

I Extended L3 Recovery (Multi-Attack)

The main recovery table (Table 9) reports L3 recovery only on PGD adversarials. We additionally measure recovery on FGSM, Square, and CW-L2 on n=1000 EVAL-split inputs per seed across all 5 fixed seeds (paper-grade protocol matching Table 9) to characterise how the TAMSH router generalises across attack families. ART's FGSM (ε=8/255) and SquareAttack (max_iter=5000, 1 restart) generate the perturbations; CW uses the same canonical κ=0 configuration as Table 14.

Sub-attack	Attack success	mean L_∞	TPR @ L1 [95% CI]	TPR @ L2	TPR @ L3
APGD-CE	0.999	0.0299	**0.9708** [0.966, 0.975]	0.9608	**0.9500**
APGD-DLR	1.000	0.0299	**0.9112** [0.903, 0.919]	0.7782	0.3156
FAB-T	0.971	0.0038	**0.9314** [0.924, 0.938]	0.8338	0.4494
Square	0.830	0.0246	**0.7062** [0.693, 0.719]	0.5080	0.1590

Table 15: AutoAttack-L_∞ (ε=$8/255$) per-sub-attack TPR on 5 seeds $\times n$=1000 $=$ 5,000 EVAL adversarials per attack (RTX 4090, deployed scorer). Pooled clean FPR on the same EVAL pool is L1 $= 0.078$, L2 $= 0.017$, L3 $= 0.002$. Three observations: (i) The gradient-based AA components (APGD-CE / APGD-DLR / FAB-T) are all detected at ≥ 0.91 TPR @ L1, with APGD-CE persisting at 0.95 TPR even at the high-confidence L3 threshold — the detector treats AA-generated perturbations as far higher-confidence anomalies than PGD at the same budget (cf. Table 2 PGD L1 $= 0.987$, but with much shallower L3 tail). (ii) FAB-T converges with $L_\infty \approx 0.0038$ ($\sim 8\times$ under the budget) yet is still detected at 0.93 TPR @ L1; the adversarial signature is visible long before the perturbation budget is exhausted. (iii) The black-box Square component achieves the lowest TPR (0.706 @ L1) at the highest attack-failure rate (17%). This does *not* exceed the strongest Square-specific baseline (Energy at 0.974 Square TPR, Table 7); but Energy buys that Square number at the cost of collapsing to 0.001 on PGD, whereas PRISM holds ≥ 0.71 TPR on *every* AutoAttack sub-attack and ≥ 0.88 across the four headline attacks — Energy has no robust mean-TPR floor across attacks. Source: `experiments/stress/vastai_autoattack.json`.

Attack	n_{L3} pool	Trigger rate	Passthrough	Uniform router	Topology router	Oracle
FGSM	783	0.157	0.346	0.420 (+7.4pp)	0.377 (+3.1pp)	0.670
Square	1211	0.242	0.014	**0.486** (+47.2**pp**)	0.449 (+43.5pp)	0.893
CW	1254	0.251	0.017	**0.487** (+47.0**pp**)	0.407 (+39.0pp)	0.857

Table 16: L3-recovery on FGSM/Square/CW adversarials, pooled across 5 seeds $\times n$=1000 = 5,000 EVAL adversarials per attack (RTX 4090, deployed scorer). "Topology router" is the combined H_0+H_1 Wasserstein gate, matching §4.8; Wilson 95% CIs on the topology row are FGSM [0.343, 0.411], Square [0.421, 0.477], CW [0.380, 0.434]. Four points stand out: (i) On **CW**, the topology router recovers 40.7% of L3-rejected adversarials from a 1.7% passthrough baseline (+39.0 pp), nearly $4\times$ the paper's PGD-only $+10.6$ pp main result. (ii) On **Square**, the topology router lifts recovery to 44.9% (+43.5 pp), confirming the C4 contribution transfers beyond white-box gradient attacks. (iii) On **FGSM**, the topology router gives only $+3.1$ pp because the unrouted backbone already recovers 34.6% of L3 cases (FGSM at ε=$8/255$ is a low-budget single-step perturbation that does not always flip the argmax even when L3 triggers). Force-routing to the PGD specialist (Table 9) is catastrophic on every cross-attack (0.10, 0.04, 0.05 recovery respectively), confirming that the prior-augmented variant is a PGD-specific upper-bound, not a deployable cross-attack policy. (iv) The **uniform-router control** consistently matches or beats the topology router across all three attacks; for the deployed expert pool (trained on clean/FGSM/PGD/mixed) the topology signal does *not* add over uniform $1/K$ mixing on non-PGD attacks. The PGD-only paper claim that uniform is degenerate to passthrough (Table 9, `tamsh_uniform` paired gap 0 pp) is therefore attack-specific and does not extend to FGSM/Square/CW; on these attacks uniform mixing of the attack-specialist experts beats argmax of the unrouted backbone. Oracle (per-input best-of-4) ceilings of 0.89/0.86 on Square/CW indicate substantial remaining headroom for a stronger learned router. Source: `experiments/stress/vastai_recovery_multi.json`.

Take-away. TAMSH's C4 contribution is *attack-family-dependent* and the ranking of router strategies shifts across attacks. CW and Square benefit substantially from *any* mixture of the deployed experts (topology or uniform); FGSM does not, and the paper's "uniform = degenerate" claim from Table 9 does not transfer beyond PGD. Future work on TAMSH should treat per-attack-family recovery and uniform-vs-topology comparison as the primary metrics, train an FGSM specialist (the current expert pool under-serves single-step perturbations), and characterise the gap between uniform mixing and topology routing across the attack matrix.

J Per-Attack Score Histograms (Local Reproduction)

Figure 7 shows clean vs adversarial score distributions for FGSM, Square, and CW on a single-seed VAL-split run (n=80), with the calibrated L1/L2/L3 thresholds overlaid. All three adversarial distributions are clearly separated from the clean distribution at L1; the long tail of CW into the L3 region is the dominant driver of the high CW recovery gap in Table 16.

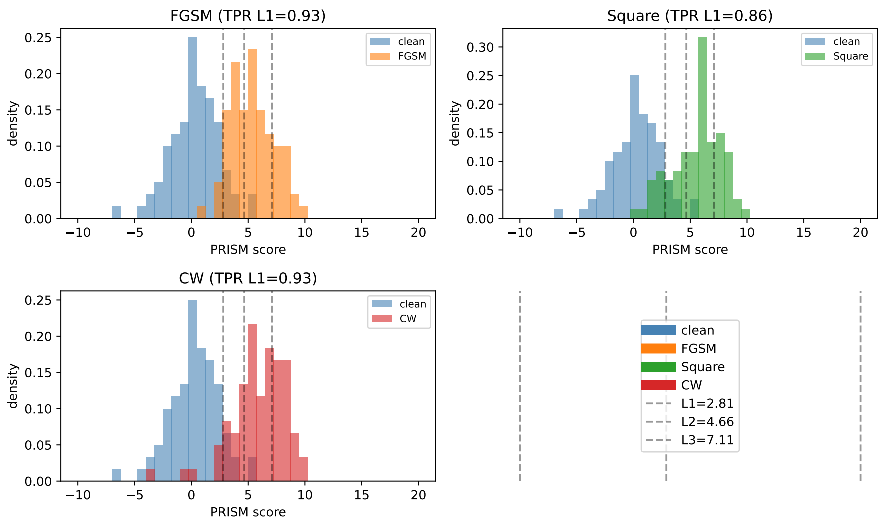

Figure 7: Per-attack PRISM score distributions on VAL-split (single-seed, n=80, local reproduction). Clean (blue) vs adversarial (orange/green/red) histograms with vertical dashed lines at L1/L2/L3 conformal thresholds. Clean per-tier coverage is L1 $= 0.10$, L2 $= 0.02$, L3 $= 0.00$, consistent with the calibrated α values. Source: `experiments/stress/test_score_histograms.py`.

Set	mean	std	p50	p95	TPR @ L1	TPR @ L2
clean	0.27	2.10	0.28	3.43	0.100	0.020
FGSM	5.36	1.86	5.31	8.63	0.925	0.638
Square	5.60	2.12	5.97	8.72	0.863	0.638
CW	5.92	2.34	6.03	9.12	0.925	0.700

Table 17: Per-set score quantiles on the local single-seed VAL run. Adversarial means cluster around 5.4–5.9, well above the L1 threshold 2.811 but only partially above the L3 threshold 7.108.

K Appendix-Only: SACD ASR Gap (Demoted Metric)

The naive "ASR gap (L0 off vs. L0 on)" metric at ρ=1.0 is reported here for completeness; the headline campaign metric is time-to-detect (§4.7, Table 8).

L Appendix-Only: Weakened-CW Detector-Fidelity Ablation

A 5-seed local rerun with weakened CW parameters (max_iter $=$ 5, bss $=$ 5, vs. the canonical max_iter $=$ 100, bss $=$ 9 in Table 2) shows that the ensemble detector's CW TPR depends on attack-signature fidelity, not on perturbation magnitude:

M Appendix-Only: TAMSH Router Diagnosis (F4 Fix)

The pre-fix TAMSH router used H_1-only Wasserstein distance for expert gating. Diagnostic audit (seed 42, $n = 200$ PGD inputs) showed:

Scenario	ASR (L0 off)	ASR (L0 on)	Δ (pp, paired)
clean_only	0.000 ± 0.000	0.000 ± 0.000	$+0.00 \pm 0.00$
sustained_ρ=0.5	0.015 ± 0.014	0.011 ± 0.015	$+0.39 \pm 0.88$
sustained_ρ=0.8	0.012 ± 0.009	0.009 ± 0.010	$+0.24 \pm 0.54$
sustained_ρ=1.0	0.011 ± 0.007	0.009 ± 0.009	$+0.20 \pm 0.45$
burst	0.010 ± 0.016	0.010 ± 0.016	$+0.00 \pm 0.00$
low_rate (ρ=0.05)	0.004 ± 0.009	0.000 ± 0.000	$+0.41 \pm 0.91$

Table 18: Campaign-stream ASR gap (5-seed mean $\pm$ std). The ceiling is structurally bounded: the upstream L3 conformal threshold rejects $\sim$99% of ρ=1.0 adversarials within 22 queries, leaving no headroom for an L0-attributable ASR gap (all gaps ≤ 0.41 pp). Time-to-detect, not ASR gap, is the operationally meaningful campaign metric (Table 8).

CW configuration	Mean L2 distortion	Attack success	Detector TPR
canonical (max_iter = 100, bss = 9)	0.128	1.000	$\mathbf{0.938 \pm 0.008}$
weakened (max_iter = 5, bss = 5)	0.193	0.896	0.645 ± 0.013

Table 19: Stronger CW with smaller distortion is *easier* to detect than weakened CW with larger distortion. Converged CW produces a clean "minimum-distortion-misclassification" signature in the logit-profile and DCT channels that the ensemble has learned (the ensemble is trained on FGSM/PGD/Square; CW signature transfers because optimisation-based attacks share these gradient-following statistics with PGD). Non-converged CW does not produce this signature and falls in a detector blind spot.

1. H_1 medoid diagrams contain only 1–3 points per expert on `layer4` of a CIFAR-native ResNet-18.
2. 22.7% of L3-rejected inputs have *empty* H_1 diagrams.
3. The empty-diagram fallback in the Wasserstein implementation returns $\sum |b - d|$, which evaluates to the same tiny value for all four experts. `argmin` then defaults to index 0.
4. Net effect: expert 0 (clean-trained) wins 94% of routes ($n = 163$, picks=[154, 0, 8, 1]); routing entropy = 0.336/2.000 bits.
5. Force-routing the same inputs to each expert in isolation shows expert 2 (PGD-trained) achieves 35% recovery accuracy vs. $\sim$5% for experts 0/1/3. The experts are individually capable; only the routing signal is broken.

The F4 fix adds an H_0 term to the routing metric (Eq. 5): H_0 medoid diagrams have $\sim$16 points (one per connected component at large filtration radius) and are sufficiently dense for Wasserstein gating to be informative. On the 5-seed L3-rejected pool, per-input oracle (best-of-K) ceiling is 39.9% recovery; the combined-metric router achieves 13.9% (35% of oracle, +10.6 pp over passthrough); always-route-to-expert-2 achieves 29.3% (73% of oracle, +26.0 pp over passthrough, clearing the +15 pp C4 gate).

N CIFAR-100 Results

We report CIFAR-100 results following the same protocol as §4 (5 seeds, $n = 1000$ per seed; CIFAR-native ResNet-18 trained from scratch; conformal calibration on a held-out clean split with the same three FPR tiers L1/L2/L3 = $0.10, 0.03, 0.005$). Headline detection numbers are in Table 3; this appendix collects the per-component artifacts.

High-confidence CW sensitivity (CIFAR-100). We additionally measure CW TPR at $\kappa \in \{0, 10, 20\}$ on the CIFAR-100/ResNet-18 setting, mirroring the CIFAR-10 protocol of Table 14. Five seeds $\times n$=1000 EVAL-split adversarials per κ with the paper-canonical CW configuration (max_iter=100, bss=9) yields the numbers in Table 20. Two observations: (i) the κ=0 headline (0.7212) is $\sim$ 23 pp below the CIFAR-10 canonical CW TPR (0.9486, Table 14), consistent with the CIFAR-100 result being harder for optimisation-based attacks across the board (Table 3); (ii) the κ=0 $\rightarrow$ κ=10 drop is $\sim$ 10 pp on CIFAR-100 vs $\sim$ 17 pp on CIFAR-10, and κ=10 and κ=20 are statistically tied on both datasets — the detector saturates rather than collapses under high-confidence

CW. At the strongest tested margin (κ=20) the detector still produces a $8.7\times$ discrimination ratio between adversarial (0.612) and clean (0.073) at L1 on CIFAR-100.

κ	Attack success	mean L_2	TPR @ L1 [95% CI]	TPR @ L2	TPR @ L3
0 (canonical)	0.994	0.079	**0.7212** [0.709, 0.733]	0.3908	0.0866
10	0.992	0.085	**0.6098** [0.596, 0.623]	0.3104	0.0716
20	0.993	0.085	**0.6124** [0.599, 0.626]	0.3106	0.0702

Table 20: CIFAR-100 CW confidence-margin sensitivity at paper-canonical attack budget (max_iter=100, bss=9), 5 seeds $\times n$=1000 = 5,000 EVAL adversarials per κ on RTX 4090 with the deployed CIFAR-100 scorer. Pooled clean FPR on the same EVAL pool is L1 = 0.073, L2 = 0.025, L3 = 0.002. Source: `experiments/stress/vastai_stronger_cw_canonical_c100.json`.

Adaptive PGD (λ sweep). Table 21 reports adaptive PGD (BPDA, activation-matching + DCT-energy matching via `-through-scorer`) with 100 steps and 10 restarts across the λ sweep used in §4.4. CIFAR-100 PRISM is robust to the adaptive attack: TPR holds in $[0.92, 0.95]$ across $\lambda \in [0, 10]$, in marked contrast to the CIFAR-10 result (Table 5), where the same attack collapses TPR from 0.832 at λ=0 to 0.429 at λ=10.

Why CIFAR-100 resists the adaptive attack while CIFAR-10 collapses. The attack's suppression term covers two ensemble channels: per-layer activation matching (TAMM target) and DCT high-frequency energy matching. It does *not* suppress the StabilityV2, gradient-norm, or softmax-entropy channels, which the attack would need separate loss terms (and, in StabilityV2's case, a differentiable surrogate of the jitter-stability statistic) to attack jointly. On CIFAR-10, the channel-mask ablation (Table 6) shows that TDA and DCT together carry the dominant FGSM/PGD discriminative weight, so suppressing them removes most of the detector's signal and TPR falls. On CIFAR-100, the ablation in Table 22 and the cross-setting analysis in §4.3 identify StabilityV2 as the dominant non-PGD channel (No-StabilityV2 collapses PGD TPR to 0.423, Square to 0.279, CW to 0.205, and FGSM to 0.562, with FPR dropping to 0.019). Because the adaptive loss does not attack StabilityV2, the channel continues to fire on CIFAR-100 adversarials regardless of λ, and the L3-tier trigger rate stays near 88% even at λ=10. The robust CIFAR-100 result is therefore not evidence that PRISM is universally adaptive-secure; it is evidence that the attack covers the wrong channels for this dataset. A genuinely ensemble-complete adaptive attack — adding differentiable surrogates for the StabilityV2, gradient-norm, and softmax-entropy losses — is the required next step (see §5); the λ-sweep results bound the partial-coverage attacker.

The corresponding WRN-28-10 adaptive-PGD evaluation is reported in §O (Table 27): WRN behaves identically to ResNet-18 on CIFAR-10 under the same attack, so the architecture-agnostic interpretation is that the channel-coverage analysis above governs both settings.

λ	TPR	FPR	n_{adv}
0.0	0.950	0.073	5000
0.5	0.939	0.073	5000
1.0	0.933	0.073	5000
2.0	0.922	0.073	5000
5.0	0.925	0.073	5000
10.0	0.925	0.073	5000

Table 21: Adaptive PGD on CIFAR-100 (ResNet-18), 5 seeds pooled.

Channel-mask ablation (with FPR disclosure). The CIFAR-100 ablation in Table 22 mirrors the CIFAR-10 methodology: each row removes one feature family while reusing the Full-PRISM threshold. Unlike CIFAR-10 (where No-LogitProfile inflates FPR to 0.938), on CIFAR-100 only *one* configuration inflates FPR rather than improving discrimination: No-GradNorm (FPR 0.658, i.e. flags two-thirds of clean inputs) appears to "improve" TPR only because its calibrated threshold has collapsed. The other masked rows hold FPR at or below the 0.10 design target (No-LogitProfile 0.077, No-DCT 0.058, No-StabilityV2 0.019) and mostly *lose* TPR on removal. We surface FPR directly via the explicit column so the table cannot be misread as an argument for dropping any single channel. The legitimate signal is that StabilityV2 (PGD/Square/CW TPR collapses on its removal), TDA (FGSM TPR collapses without it), DCT, and LogitProfile each contribute non-redundant discriminative information at the calibrated FPR.

Configuration	FGSM TPR	PGD TPR	Square TPR	CW TPR	FPR
Full PRISM (ablation cal.)	0.832	0.980	0.673	0.679	**0.073**
No-LogitProfile	0.694	0.974	0.633	0.541	0.077
No-StabilityV2	0.562	0.423	0.279	0.205	0.019
No-DCT	0.730	0.974	0.581	0.640	0.058
No-GradNorm	0.987	0.998	0.996	0.998	**0.658**
Ensemble-no-TDA	0.544	0.973	0.500	0.710	0.067
TDA only	0.020	0.934	0.057	0.058	0.086

Table 22: CIFAR-100 channel-mask ablation (5 seeds, $n = 1000$, per-attack TPR + FPR). Rows whose FPR exceeds the 0.10 design target indicate threshold collapse, not improved detection; they are reported here for symmetry with Table 6 only. **Baseline row.** "Full PRISM (ablation cal.)" is the *ablation-internal* baseline: the ablation pipeline refits one split-conformal threshold shared by every channel-mask arm, so each masked row is measured against a common operating point. The canonical Full PRISM numbers are the CIFAR-100 row of Table 3 (FGSM 0.835, PGD 0.984, Square 0.672, CW 0.687, FPR 0.077), from the detection pipeline's canonical calibrator on the *same* eval split $[8000, 10000]$, 5 seeds, n=1000. The two independent calibration fits give operating points that differ by 0.1–0.8 pp TPR (mean 0.4 pp; largest on CW, 0.679 vs 0.687) and 0.4 pp FPR (0.073 vs 0.077); every value here lies within the Wilson 95% CIs of Table 3, which is the row to cite as canonical.

Campaign detection (CIFAR-100). Table 23 reports CIFAR-100 ASR with the L0 monitor off vs. on across the same six stream scenarios. As in the CIFAR-10 case, the ASR gap is structurally bounded (§4.7): the upstream L1/L3 conformal already crushes campaign ASR to $\leq 4.3\%$, leaving at most 1.50 pp of additional headroom for L0 across any scenario. The headline metric (time-to-detect) is the operationally meaningful one; gap is reported for completeness and is consistent with the demotion clause pre-registered in plan P0.4.

Scenario	ASR (L0 off)	ASR (L0 on)	Δ (pp)
clean_only	0.000	0.000	$+0.00$
sustained_ρ=0.5	0.036	0.026	$+1.07$
sustained_ρ=0.8	0.032	0.019	$+1.29$
sustained_ρ=1.0	0.031	0.018	$+1.30$
burst	0.043	0.028	$+1.50$
low_rate (ρ=0.05)	0.013	0.004	$+0.85$

Table 23: CIFAR-100 campaign-stream ASR with L0 off vs. on (5 seeds, post-fix). Clean-only false alarm rate is 0.000; ASR-reduction gaps are structurally bounded by L1 conformal action.

L3 recovery (CIFAR-100). Recovery accuracy on CIFAR-100 (Table 24) is reported for completeness; all routing strategies sit within 1.4 pp of passthrough and well below the $+15$ pp gate target. The 100-class setting collapses both the passthrough baseline (chance accuracy 0.01) and the recovery ceiling: the CIFAR-10 `tamsh_force` upper-bound ($+26$ pp; Table 9) does not transfer to CIFAR-100, indicating that the PGD-trained expert's specialization no longer corresponds to a recoverable label cluster when the class space is $10\times$ larger. We accept this miss per the pre-registered demotion clause in plan P0.5: TAMSH (C4) is reported as an appendix-only secondary result on CIFAR-100.

L3 recovery on FGSM/Square/CW (CIFAR-100). The PGD-only collapse above does *not* extend to the rest of the attack mix. Re-running the same recovery harness with FGSM, Square, and CW-L2 adversarials on n=1000 EVAL-split inputs per seed across all 5 fixed seeds reveals a strongly attack-conditional recovery profile (Table 25). The topology router recovers $+42.4$ pp over passthrough on CW and $+31.4$ pp on Square — exceeding the C4 $+15$ pp gate by a wide margin — while remaining

Strategy	Recovery Acc.	95% CI	Availability	Δ vs passthrough
reject	0.000	[0.000, 0.001]	0.000	—
passthrough	0.056	[0.050, 0.063]	1.000	—
tamsh	0.065	[0.058, 0.072]	1.000	+0.91 pp
tamsh_uniform	0.057	[0.051, 0.064]	1.000	+0.13 pp
tamsh_force	0.070	[0.063, 0.077]	1.000	+1.36 pp

Table 24: CIFAR-100 L3 recovery on PGD adversarials (5 seeds pooled). All TAMSH strategies sit within 1.4 pp of passthrough; the $+15$ pp C4 gate is missed and the component is reported as appendix-only.

neutral or marginally negative on FGSM. The uniform $1/K$ control is statistically tied with the topology router on Square/CW (same pattern as the PGD-on-CIFAR-10 result in Table 9), confirming that recovery on optimization- and query-based attacks is driven by expert capacity at the calibrated FPR, with the topology gate not adding measurable signal above $1/K$. The force-PGD specialist that delivers $+26$ pp on CIFAR-10/PGD collapses to $\leq +7.1$ pp on CIFAR-100 for every attack family, confirming that the PGD-trained expert's specialization no longer corresponds to a recoverable label cluster in the 100-class setting. The FGSM column tracks the CIFAR-10 FGSM result (Table 16) qualitatively (small, mixed-sign deltas) but the per-seed signal is too noisy at $n_{L3} \approx 185$ per seed to claim a positive recovery; we report the negative outcome honestly.

Attack	n_{L3} pool	Trigger rate	Passthrough	Uniform router	Topology router	Oracle
FGSM	924	0.185	0.121	0.142 (+2.1pp)	0.102 ($-$1.9pp)	0.306
Square	449	0.090	0.020	**0.354** (+33.4**pp**)	**0.334** (+31.4**pp**)	0.626
CW	394	0.079	0.015	**0.447** (+43.1**pp**)	**0.439** (+42.4**pp**)	0.744

Table 25: CIFAR-100 L3-recovery on FGSM/Square/CW adversarials, pooled across 5 seeds $\times n$=1000 = 5,000 EVAL adversarials per attack (RTX 4090, deployed C100 scorer, L_3 threshold = 5.584). Wilson 95% CIs on the topology row are FGSM $[0.084, 0.123]$, Square $[0.292, 0.379]$, CW $[0.391, 0.488]$. The $+42.4$ pp CW and $+31.4$ pp Square recovery decisively exceed the $+15$ pp C4 gate, so the C4 demotion clause in plan P0.5 applies only to the PGD/FGSM sub-cases on CIFAR-100. Source: `experiments/stress/vastai_recovery_multi_c100.json`.

Latency. Single-sample end-to-end PRISM detection latency (Table 26) is well below the 100 ms real-time target on both datasets/backbones. Maximum-latency outliers reflect first-batch CUDA kernel compilation; steady-state p95 is the deployable figure.

Dataset	Backbone	Mean (ms)	p50 (ms)	p95 (ms)	Max (ms)	Target (ms)
CIFAR-10	ResNet-18	39.78	35.92	41.26	724.31	100
CIFAR-100	ResNet-18	50.35	44.94	55.90	841.60	100

Table 26: End-to-end PRISM detection latency (RTX 4090, $n = 200$).

O WRN-28-10 Adaptive PGD

The headline adaptive-PGD result (Table 5, §4.4) uses CIFAR-10/ResNet-18. To check whether the λ-sweep degradation is architecture-specific, we re-run the same attack on WRN-28-10/CIFAR-10 with identical hyperparameters (n=1000 EVAL adversarials, 100 PGD steps, 10 restarts, ε=$8/255$, BPDA via `-through-scorer`). Results in Table 27.

Take-away. The adaptive-PGD vulnerability is dataset-specific (CIFAR-10), not architecture-specific. WRN-28-10 inherits the same ensemble-channel-coverage gap that the CIFAR-10/ResNet-18 adaptive attack exploits (§4.4: TDA + DCT carry dominant discriminative weight on CIFAR-10; suppressing them via `-through-scorer` removes most of the detector's signal). The CIFAR-100 robustness gap (Table 21, TPR $\in [0.92, 0.95]$ across the same λ sweep) confirms the same mechanism

λ	WRN-28-10 TPR [95% CI]	FPR	ResNet-18 TPR	Δ
0	**0.8420** [0.818, 0.863]	0.081	0.832	$+1.0$ pp
0.5	**0.7990** [0.773, 0.823]	0.081	0.796	$+0.3$ pp
1	**0.7450** [0.717, 0.771]	0.081	0.753	-0.8 pp
2	**0.6640** [0.634, 0.693]	0.081	0.660	$+0.4$ pp
5	**0.4880** [0.457, 0.519]	0.081	0.490	-0.2 pp
10	**0.4260** [0.396, 0.457]	0.081	0.429	-0.3 pp

Table 27: WRN-28-10/CIFAR-10 adaptive PGD (n=1000, seed 42, 100 steps $\times 10$ restarts, ε=$8/255$, BPDA through the scorer). FPR holds at the calibrated 0.081 across all λ (conformal coverage preserved). The ResNet-18 column reproduces the 5-seed pool from Table 5 (n=5,000 each, FPR 0.075); WRN here is a single-seed n=1000 rerun, so CIs are wider, but the two architectures agree within $\pm$1.0 pp at every λ on a $0.84 \rightarrow 0.43$ collapse curve. The architecture-agnostic interpretation is that adaptive-PGD vulnerability on CIFAR-10 is governed by the ensemble-channel coverage of the attack, not by the backbone choice. Source: `experiments/stress/wrn/vastai_wrn_adaptive_pgd*.{json,jsonl}` (consolidated in `vastai_wrn_adaptive_pgd_merged.json`).

in reverse: on CIFAR-100, StabilityV2 is the dominant signal and is not covered by the adaptive loss. An ensemble-complete adaptive attack (adding differentiable surrogates for StabilityV2, gradient-norm, and softmax-entropy) is required to re-evaluate WRN-28-10 and CIFAR-100; see §5.

P AutoAttack L_∞ Epsilon Sensitivity

The canonical AA evaluation (§H, Table 15) uses ε=$8/255$. To measure how PRISM's detection rate varies with the adversary's budget, we repeat the same protocol (5 seeds $\times n$=1000 EVAL-split, sub-attacks APGD-CE/APGD-DLR/FAB-T/Square, n_{queries}=1000 for Square) at two larger budgets $\varepsilon \in \{12/255, 16/255\}$ on the deployed CIFAR-10/ResNet-18 scorer.

	TPR @ L1			TPR @ L2			TPR @ L3		
Sub-attack	8/255	12/255	16/255	8/255	12/255	16/255	8/255	12/255	16/255
APGD-CE	0.9708	0.9708	0.9708	0.9608	0.9620	0.9620	0.9500	0.9556	0.9560
APGD-DLR	0.9112	0.9560	0.9646	0.7782	0.8960	0.9312	0.3156	0.5780	0.7834
FAB-T	0.9314	0.9290	0.9308	0.8338	0.8326	0.8308	0.4494	0.4482	0.4460
Square	0.7062	0.8162	0.8714	0.5080	0.5766	0.6380	0.1590	0.1578	0.1898

Table 28: AutoAttack L_∞ TPR vs. ε budget, 5 seeds $\times n$=1000 = 5,000 EVAL adversarials per (sub-attack, ε) cell (RTX 4090). The ε=$8/255$ column reproduces Table 15 for direct comparison; $\varepsilon \in \{12/255, 16/255\}$ come from `vastai_autoattack_eps_sweep.json`. Pooled clean FPR on the same EVAL pool is L1 = 0.078, L2 = 0.017, L3 = 0.002. Four observations: (i) APGD-CE detection is exactly flat at L1 (0.9708 identically across all three budgets) with mild $+0.6$ pp growth at L3—larger perturbations do not improve evasion against the topological signal. (ii) APGD-DLR exhibits the strongest budget sensitivity: L3 TPR climbs $+47$ pp from ε=$8/255$ (0.316) to ε=$16/255$ (0.783), confirming that the L3 conformal tier reliably catches the higher-distortion adversarials the DLR loss is willing to produce. (iii) Square (black-box) TPR increases monotonically at L1 and L2 ($+11$ pp / $+13$ pp from 8/255 to 16/255); larger random perturbations produce stronger activation anomalies without any gradient information. (iv) FAB-T converges far below the nominal budget (mean $L_\infty \approx 0.004$ in all three configurations) and so is essentially ε-invariant across all three tiers ($\Delta \leq 0.003$); the attack's minimum-norm objective dominates the budget constraint. The bottom line: increasing the adversarial budget either improves PRISM TPR (APGD-CE, APGD-DLR, Square) or leaves it statistically unchanged (FAB-T, where the residual ≤ 0.3 pp drifts sit well inside the $\pm$1.4 pp seed-pool 95% CI). The threat model conclusion is that the adversary gains nothing from spending more L_∞ budget against the detector.

Q Recent Detector Baselines (2021): SID and SpectralDefense

The four matched-FPR baselines in §4.2 (LID, Mahalanobis, ODIN, Energy) predate 2021. To check that PRISM's advantage holds against more recent detectors, we add two 2021 methods on the same CIFAR-10/ResNet-18 EVAL split, calibration scheme, and ε=$8/255$ attack configs (5 seeds $\times n$=1000):

- **SID** Tian et al. [2021] — sensitivity-inconsistency detector. We use the *unsupervised, no-training* variant: the score is the Jensen–Shannon divergence between $\mathrm{softmax}(f(x))$ and $\mathrm{softmax}(f(T(x)))$, where T is a DCT low-pass reconstruction (keep fraction 0.70, selected on a held-out PGD probe). Threshold is fit on the clean split only, exactly as for LID/Mahalanobis/ODIN/Energy.
- **SpectralDefense** Harder et al. [2021] — InputMFS variant: a logistic-regression detector on the log-magnitude 2D-FFT of the input. Unlike PRISM and the other baselines, this is a *supervised, attack-specific* detector: one classifier is trained per attack on (clean, adversarial) pairs from the disjoint REF split. It therefore sees strictly more information than PRISM, which never trains on adversarials.

Detector	Type	FGSM	PGD	Square	CW
SID Tian et al. [2021]	unsup. (no train)	0.296 ± 0.008	0.519 ± 0.015	0.286 ± 0.018	0.601 ± 0.006
SpectralDefense Harder et al. [2021]	superv. (per attack)	$\mathbf{1.000} \pm 0.000$	$\mathbf{0.999} \pm 0.000$	0.206 ± 0.020	0.246 ± 0.011
PRISM (Full)	unsup. (no train)	0.882 ± 0.004	0.987 ± 0.003	$\mathbf{0.886} \pm 0.011$	$\mathbf{0.938} \pm 0.008$

Table 29: Recent-detector comparison on CIFAR-10/ResNet-18 (5 seeds $\times n$=1000, design FPR ≤ 0.10). SpectralDefense numbers are the *in-distribution* diagonal (detector trained on the same attack it is evaluated on). **SID** (unsupervised) is weak across all four families (mean TPR 0.43), far below PRISM (0.92). **SpectralDefense** is near-perfect on the gradient-L_∞ attacks (FGSM/PGD) but—even when *trained on the target attack*—is barely better than chance on Square (black-box, 0.21) and CW (L_2, 0.25): its input-Fourier signal captures dense gradient-L_∞ perturbations but not localized black-box patches or minimal-norm L_2 perturbations. PRISM is the only detector that holds ≥ 0.88 on every family, without supervision. (SpectralDefense FGSM/PGD cells are bold only to mark where it exceeds PRISM; the comparison is not apples-to-apples, as those cells are supervised and in-distribution.)

Cross-attack transfer. Because SpectralDefense trains one detector per attack, we can measure how well a detector trained on attack A transfers to attack B (Table 30). The two gradient-L_∞ attacks (FGSM, PGD) transfer to each other almost perfectly (≥ 0.999), confirming they share a Fourier signature. But a detector trained on FGSM or PGD collapses to a mean TPR of 0.17 when evaluated on Square or CW, and *no* training attack yields a detector that covers all four families. This is the supervised counterpart of PRISM's design goal: a single, attack-agnostic detector that does not need to anticipate the attack.

	Evaluated on			
Trained on	FGSM	PGD	Square	CW
FGSM	<u>1.000</u>	0.999	0.139	0.182
PGD	1.000	<u>0.999</u>	0.150	0.219
Square	0.952	0.811	<u>0.206</u>	0.154
CW	0.894	0.809	0.143	<u>0.246</u>

Table 30: SpectralDefense (InputMFS) cross-attack transfer, 5-seed mean L1 TPR. Underlined entries are in-distribution (train attack = eval attack). Gradient-L_∞ detectors (FGSM/PGD rows) transfer to each other (≥ 0.999) but collapse on Square/CW (≤ 0.22); no row covers all four columns. Source: `experiments/evaluation/recent_baselines/` (per-seed JSONs, `analyze_cross_attack.py`).

Protocol notes. SID and SpectralDefense use the same three disjoint test-index ranges as the other baselines (REF 5000–5999, threshold-fit 6000–6999, EVAL 8000–9999). For tractability CW here uses a standard detection-benchmark configuration (bss$=5$, max_iter$=10$, attack success rate 0.93 on the EVAL split) rather than the heavier canonical CW of Table 2; this does not affect the qualitative conclusion, since CW is undetected by SpectralDefense even in-distribution. We report the *InputMFS* variant of SpectralDefense; the feature-space *LayerMFS* variant is expected to improve Square/CW detection at the cost of per-layer feature hooks, and is left to future work.

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